

6

Frictional Heating and Contact Temperatures

- 6.1 [Surface Temperatures and Their Significance](#)
- 6.2 [Surface Temperature Analysis](#)
 - Analytical Methods for Flash Temperature Rise Calculations •
 - Numerical Methods for Surface Temperature Determination
- 6.3 [Surface Temperature Measurement](#)
 - Thermocouples, Thermistors, and Related Temperature
 - Sensors • Radiation Detection Techniques • Ex Post Facto
 - Methods

Francis E. Kennedy
Dartmouth College

6.1 Surface Temperatures and Their Significance

Friction occurs whenever two solid bodies slide against each other. It takes place by a variety of mechanisms in and around the real area of contact between the sliding or rolling/sliding bodies. It is through frictional processes that velocity differences between the bodies are accommodated. It is also through these processes that mechanical energy is transformed into internal energy or heat, which causes the temperature of the sliding bodies to increase. The exact mechanism by which this energy transformation occurs may vary from one sliding situation to another, and the exact location of that transformation is usually not known for certain. It is known that solid friction and related frictional processes, including frictional heating, are concentrated within the real area of contact between two bodies in relative motion. Some investigators contend that these processes occur by atomic-scale interactions within the top several atomic layers on the contacting surfaces (Landman et al., 1993), while others believe that most energy dissipation occurs in the bulk solid beneath the contact region by plastic deformation processes (Rigney and Hirth, 1979). Experimental work has shown that at least 95% of the energy dissipation occurs within the top 5 μm of the contacting bodies (Kennedy, 1982). Although there may be disagreement about the exact mechanism of the energy transformation, most tribologists agree that nearly all of the energy dissipated in frictional contacts is transformed into heat (Uetz and Föhl, 1978). This energy dissipation, called *frictional heating*, is responsible for increases in the temperatures of the sliding bodies, especially within the contact region on their sliding surfaces where the temperatures are highest. For the purposes of this discussion, it will be assumed that all frictional energy is dissipated as heat which is conducted into the contacting bodies at the actual contact interface.

Frictional heating and the resulting contact temperatures can have an important influence on the tribological behavior and failure of sliding components. Surface and near-surface temperatures can become high enough to cause changes in the structure and properties of the sliding materials, oxidation of the surface, and possibly even melting of the contacting solids. These temperature increases can often be responsible for changes in the friction and wear behavior of the material or the behavior of any

lubricant present in the contact. Among the most important effects of frictional heating on tribological processes are the following:

- Sliding friction of materials with low melting points is often dominated by frictional heating effects. The low sliding friction of ice and snow is due to the presence of a thin lubricating layer of meltwater which results from frictional heating of contacting ice crystals (Bowden and Hughes, 1939; Oksanen and Keinonen, 1982; Akkok et al., 1987). The low friction brought about by meltwater lubrication is critical for winter sports such as skiing, ice skating, and bobsledding. The temperature rise at the contacting ice asperities may not be sufficient to cause melting at very low sliding speeds; as a result the friction coefficient of ice and snow at very slow velocities may be as high as 0.6 to 0.8, but as soon as higher sliding velocities cause sufficient melting of the surface to produce a lubricating layer, the friction coefficient drops below 0.1 (Kennedy et al., 1999).
- Even metallic components can have contact temperatures which are sufficiently high to melt the sliding surfaces within the real area of contact if the sliding speeds are high enough. As is the case with ice, this results in a thin lubricating layer of molten material which lowers the friction significantly and increases wear (Montgomery, 1976; Carignan and Rabinowicz, 1980). Such a condition occurs with rocket sleds and with projectiles traveling in gun barrels.
- Elastomers and polymers also have friction and wear behavior which is significantly affected by interface temperatures. This regime of tribological behavior has been called the “thermal control regime” (Ettles, 1986). Frictional heating can cause surface temperatures to reach the melting or softening temperature of thermoplastic polymers, and this results in a drastic change in the friction and wear behavior of the polymer (Lancaster, 1971; Kennedy and Tian, 1994). In fact, Lancaster showed that the “PV limit,” which is often used in the design of dry plastic bearings, is in reality a “critical surface temperature limit”; i.e., the combination of contact pressure and sliding velocity causes the surface temperature to reach the critical temperature of the material (Lancaster, 1971). Even if the surface temperature does not reach the critical temperature, the viscoelastic behavior of the polymer or elastomer can be significantly affected and the resulting friction can be altered (Ettles and Shen, 1988). Design methodology has been developed for polymer bearings based on the avoidance of surface temperatures which could be detrimental to their tribological performance (Floquet et al., 1977).
- In normal circumstances, most metallic tribological components do not have contact temperatures which approach their melting or softening point, but the temperatures can still have a very major effect on their tribological behavior.
 - The phenomenon of galling or seizing of metals results from a combination of contact temperature and contact pressure at asperity junctions sufficient to cause microwelding of the two surfaces at those points. A galling criterion based on thermal considerations was developed by Ling and Saibel (1957/58).
 - The oxidation which occurs on sliding metallic surfaces exposed to air or to oxygen-containing lubricants is of great practical importance. When the oxide film is coherent and well bonded to the surface, it is beneficial because it prevents metal-metal contact and thus lowers wear and friction. When the oxide film is easily removed from the surface, however, oxidation is detrimental because it promotes wear by third-body abrasion by hard oxide particles. The subject of oxidation in sliding components and its relation to surface temperatures was reviewed by Quinn (1983) and Quinn and Winer (1985).
 - Contact temperatures and the resulting thermal stresses can play an important role in wear of sliding metallic components. The fact that temperature gradients around the contacts are very large can be responsible for softening and shear failure of the near-surface layer of the material in many situations (Rozeanu and Pnueli, 1978). The thermomechanical stress field around a sliding contact can be responsible for wear of the contacting materials, and it can be an

important wear mechanism for both ceramics and metals; such wear can be modeled by the “thermomechanical wear theory” (Ting, 1988).

- “Wear mechanism maps” have been developed in recent years to show graphically the transitions between the different mechanisms of dry sliding wear of metals (Lim and Ashby, 1987). Many of the wear mechanisms, such as mild oxidative wear, severe oxidative wear, and melt wear are very much affected by temperature, and the transitions can be dominated by contact temperature effects. As a result, the wear mechanism maps are, to a large extent, based on surface temperature maps (Lim and Ashby, 1987; Ashby et al., 1991).
- Owing to the high hardness and low thermal conductivity of many ceramics, the real contact areas of sliding ceramic components are often very small and very hot (Griffioen et al., 1986). The high contact temperatures and large temperature gradients can be responsible for large thermomechanical stresses which cause thermocracking and wear of the sliding ceramic surfaces. These phenomena are particularly important for ceramics such as zirconia which are susceptible to thermoelastic instability and thermocracking (Lee et al., 1993). Contact temperatures may play an even greater role in wear transitions for ceramics than in those for metals (Hsu and Shen, 1996).
- Scuffing or scoring is an important failure mode for lubricated sliding or rolling/sliding components such as gears or cams. Most models for scuffing failure are based on a critical contact temperature which causes scuffing by either weakening the lubricant film so it is unable to support the load (Blok, 1937; Dyson, 1975) or increasing the shear stress in the film to a limiting value (Jacobsen, 1990). The effect of temperature on the lubricant and its additives is also important (Enthoven et al., 1993).
- The effectiveness of boundary lubrication is strongly dependent on contact temperature. Friction and wear with lubricants containing physically or chemically adsorbed additives deteriorate rapidly when the contact temperature reaches a critical value at which the additive molecules desorb (Fein et al., 1959). The role of frictional heat in the lubricant transition has been well established (Ettles et al., 1994). Lubricants containing extreme pressure (EP) additives rely on high contact temperatures to promote the formation of protective films on the contact surfaces (Spikes and Cameron, 1974).
- The thermal deformations around frictionally heated contacts can give rise to the phenomenon known as thermoelastic instability (TEI). The disturbances in contact geometry, temperature, and stress which accompany TEI can have a significant effect on the performance of brakes (Dow, 1980; Barber et al., 1986), mechanical face seals (Banerjee and Burton, 1979; Kennedy and Karpe, 1982), electrical contacts (Dow and Kannel, 1982), and gas path seals (Kennedy, 1984). The development of TEI involves the interaction of frictional heating, surface deformation, and wear, and its avoidance requires an understanding of those three phenomena and their interaction.
- In addition to the effects noted above, excessive surface temperatures can contribute to the operational failure of many tribological components. One important example is magnetic storage devices. Failure of magnetic tape systems is frequently influenced by the temperature of the head/tape interface (Bhushan, 1987b), and magnetic disk storage systems are subject to degradation of protective coatings and/or lubricant, owing to high temperatures at the head/disk interface (Bhushan, 1992).

The ability to predict and measure the surface temperatures of actual contacting bodies is important if failure of tribological components is to be avoided. In addition, frictional heating has such an important influence on the tribological behavior of so many sliding systems that all tribotests must be designed with thermal considerations in mind (Floquet, 1983), and frictional heating must be considered in interpreting the results of tribotests.

Before describing the methods used for surface temperature determination, it may be useful to review the geometric and temporal conditions under which the contact temperatures occur. As is shown in [Figure 6.1](#), there are three levels of temperature in sliding contacts. The highest contact temperatures,

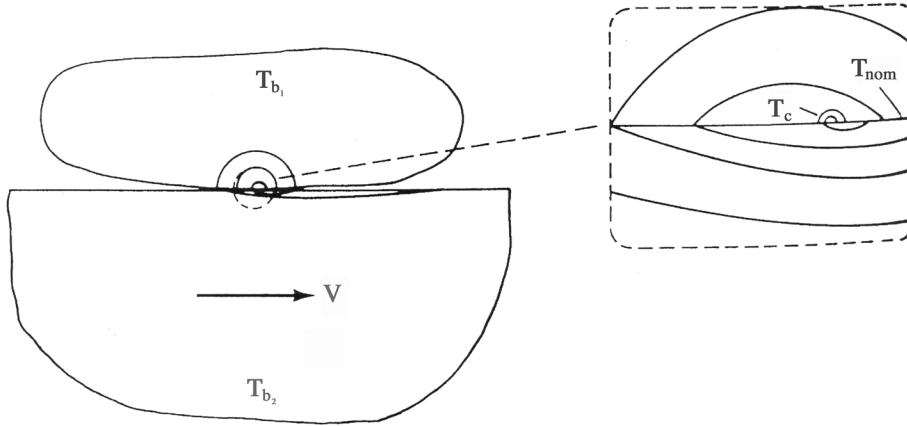


FIGURE 6.1 Schematic diagram of temperature distribution (isotherms) around sliding contact.

T_c occur at the small (perhaps on the order of 10 μm diameter) contact spots between surface roughness peaks or asperities on the sliding surfaces. These temperatures can be very high (over 1000°C in some cases) but last only as long as the two asperities are in contact. This could be less than 10 μs . The asperity contacts are often confined to a small portion of the surface of the bodies, which could be called the nominal contact patch. An example of this is a typical elliptical Hertzian contact area of several hundred μm length between two contacting gear teeth. At any instant, there are usually several short-duration flash temperature rises (ΔT_f) at the various asperity contact spots within a nominal contact patch. The integrated (in space and time) average of the temperatures of all points within the contact patch could be called the nominal (or mean) contact temperature (T_{nom}). The nominal contact temperature can be over 500°C for severe sliding cases, such as in brakes, but is usually much lower. The temperature diminishes as one moves away from the contact patch, and it generally decreases to a rather modest bulk volumetric temperature (T_b) several mm into the contact bodies. That temperature is generally less than 100°C. The total contact temperature (T_c) at a given point is given by the total of the three contributions:

$$T_c = T_b + \Delta T_{\text{nom}} + \Delta T_f \quad (6.1)$$

In the remainder of this chapter, methods will be discussed for determining the interfacial contact temperatures, T_c , in tribological systems. This temperature determination can be made by either analytical prediction or experimental measurement, and both of these methods will be discussed.

6.2 Surface Temperature Analysis

Consider a general two-body sliding (or rolling/sliding) contact in which Body 1 is moving with velocity V_1 relative to the contact area and Body 2 is moving with velocity V_2 relative to the same contact area. The rate of total energy dissipated in the sliding contact is determined by the friction force and the relative sliding velocity. If it is assumed that all of this energy is dissipated as heat on the sliding surfaces within the real area of contact, then the rate of heat generated per unit area of contact, q_{total} , is given by:

$$q_{\text{total}} = \mu p U \quad (6.2)$$

where μ is the coefficient of friction

p is the contact pressure (which may vary within the contact area)

U is the relative sliding velocity = $|V_2 - V_1|$

Fourier's law for heat conduction in an isotropic solid which is moving with velocity $\mathbf{V}$ may be written:

$$\nabla \cdot k \nabla T + \dot{Q} = \rho C \frac{DT}{Dt} = \rho C \left(\frac{\partial T}{\partial t} + \mathbf{V} \cdot \nabla T \right) \quad (6.3)$$

where $\dot{Q}$ is internal heat generation rate per unit volume, k is thermal conductivity, ρ is density, and C is specific heat.

If there is no internal heat generation and if k is uniform and constant:

$$k \nabla^2 T = \rho C \left(\frac{\partial T}{\partial t} + \mathbf{V} \cdot \nabla T \right) \quad (6.4)$$

or

$$\nabla^2 T = \frac{1}{\kappa} \frac{DT}{Dt} \quad (6.5)$$

where $\kappa = \frac{k}{\rho C}$ = thermal diffusivity.

The problem in surface temperature analysis is to determine the solution to (Equation 6.5) subject to boundary conditions which include the heat generation (Equation 6.2) at the contact interface and other thermal boundary conditions suitable for the operating conditions and geometry of the contacting solid bodies. Both analytical and numerical methods have been used to solve for the surface temperatures resulting from frictional heating.

6.2.1 Analytical Methods for Flash Temperature Rise Calculations

Most surface temperature analyses have been based on the pioneering work of Blok (1937) and Jaeger (1942), both of whom used heat source methods. Similar methods were later used by many other investigators, such as Kuhlmann-Wilsdorf (1987) and others. Heat source methods are most useful for determining the flash temperature rise component of contact temperature.

6.2.1.1 Flash Temperature Rise due to Stationary Heat Source on a Stationary Body

6.2.1.1.1 Continuous Point Heat Source on Surface of Stationary Semi-infinite Body

Assume that a heat source with constant heat supply rate Q is activated at time $t = 0$ at a point $x = x'$, $y = y'$, $z = 0$, as shown in [Figure 6.2](#). Let the surface $z = 0$ be insulated except at the heat source. It is shown in Carslaw and Jaeger (1959) that the solution for this transient problem is given by

$$\Delta T(x, y, z, t) = \frac{Q/\rho C}{4(\pi \kappa)^{3/2}} \int_{t'=0}^t \frac{e^{-\left(r^2/4\kappa(t-t')\right)}}{(t-t')^{3/2}} dt' \quad (6.6)$$

where $r = [(x - x')^2 + (y - y')^2 + z^2]^{1/2}$ is the radius from the heat source to the point of interest. Using the definition of the complementary error function

$$\operatorname{erfc}(X) = \frac{2}{\sqrt{\pi}} \int_X^\infty e^{-X^2} dX$$

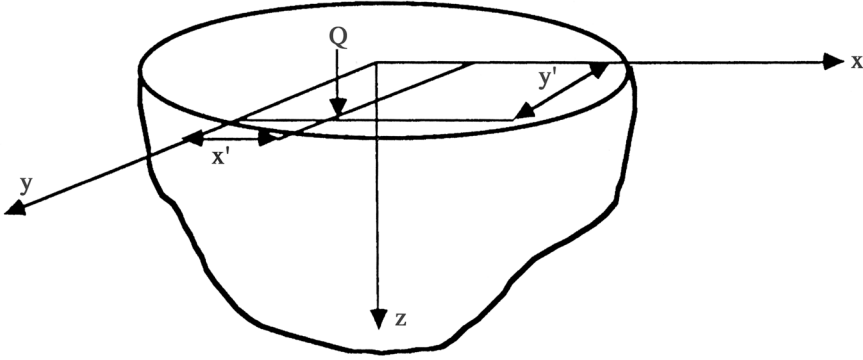


FIGURE 6.2 Point source of heat on a stationary half-space.

Equation 6.6 becomes

$$\Delta T = \frac{Q/\rho C}{2\pi\kappa r} \operatorname{erfc}\left(\frac{r}{\sqrt{4\kappa t}}\right) \quad (6.7)$$

The complementary error function $\operatorname{erfc}(X)$ shows a variation of the form shown in Figure 6.3. Therefore, $\operatorname{erfc}(0) = 1$, so the steady-state temperature (for $t \rightarrow \infty$ or $X \rightarrow 0$) is given by

$$\Delta T_{ss} = \frac{Q/\rho C}{2\pi\kappa r} = \frac{Q}{2\pi\kappa r} \quad (6.8)$$

Note: this solution is not valid at the origin, where $r = 0$. This implies that the surface temperature becomes extremely high at the concentrated point heat source. In actuality, though, the heat input cannot be concentrated at an infinitesimally small point. It must be distributed over a finite area (the real area of contact).

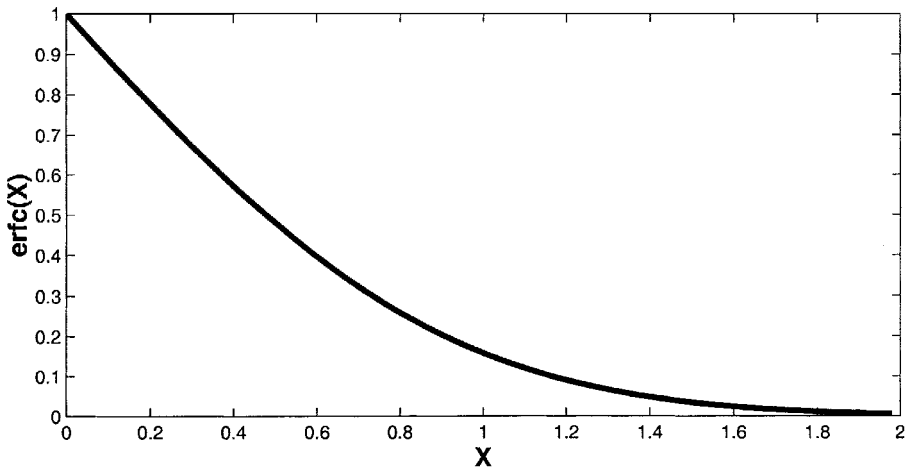


FIGURE 6.3 Complementary error function, $\operatorname{erfc}(X)$.

6.2.1.1.2 Distribution of Continuous Heat Flux on Surface of Stationary Half-Space

Let q equal the rate of heat supply per unit area. We can treat a distributed heat flux as a collection of continuous point heat sources.

A heat source $Q = q \, dx' \, dy'$ can be considered to act at point x', y' . The steady-state temperature at point $P(x, y, z)$ due to that source is found, using Equation 6.8, to be

$$dT = \frac{q \, dx' \, dy'}{2\pi k \left[(x - x')^2 + (y - y')^2 + z^2 \right]^{1/2}} \quad (6.9)$$

By superposition, the steady state temperature rise at P due to all heat sources is the integral

$$\Delta T = \iint_{A'} \frac{q \, dx' \, dy'}{2\pi k \sqrt{(x - x')^2 + (y - y')^2 + z^2}} \quad (6.10)$$

where the heat flux q may be a function x' and y' , and A' is the area over which $q(x', y')$ is distributed. Similarly, the transient temperature rise at P due to q is found (from Equation 6.7) to be

$$\Delta T = \iint_{A'} \frac{q}{2\pi k \sqrt{(x - x')^2 + (y - y')^2 + z^2}} \operatorname{erfc} \left(\frac{(x - x')^2 + (y - y')^2 + z^2}{4\kappa t} \right)^{1/2} dx' \, dy' \quad (6.11)$$

6.2.1.1.2.1 Constant Heat Supply Over Entire Surface $z = 0$ ($-\infty < x' < \infty$, $-\infty < y' < \infty$)

If q is constant (uniform heat flux) over the entire surface of a half space, Equation 6.11 integrates to give

$$\Delta T = \frac{2q}{k} \left\{ \left(\frac{\kappa t}{\pi} \right)^{1/2} e^{-z^2/4\kappa t} - \frac{z}{2} \operatorname{erfc} \frac{z}{2\sqrt{\kappa t}} \right\} \quad (6.12)$$

This is the case of linear heat transfer in a rod due to a heat flux q at the end (Carslaw and Jaeger, 1959). The temperature rise at the heated end, $z = 0$, is given by

$$\Delta T = \frac{2q}{k} \left(\frac{\kappa t}{\pi} \right)^{1/2} = 2q \left(\frac{t}{\pi \rho C k} \right)^{1/2} \quad (6.13)$$

From Equation 6.13 we can see that the surface temperature rise for this case of linear heat transfer is directly proportional to the heat flux and that an increase in either thermal conductivity or heat capacity will lead to a decrease in surface temperature.

6.2.1.1.2.2 Constant Heat Supply on Infinite Strip $-b \leq x' \leq b$, $-\infty \leq y' \leq \infty$

For this case of a band heat source on the surface of a semi-infinite solid, the temperature distribution on the surface $z = 0$ is given by (Carslaw and Jaeger, 1959)

$$\Delta T = \frac{q}{k} \left(\frac{\kappa t}{\pi} \right)^{1/2} \left\{ \operatorname{erf} \frac{b+x}{2\sqrt{\kappa t}} + \operatorname{erf} \frac{b-x}{2\sqrt{\kappa t}} - \frac{b+x}{2\sqrt{\pi \kappa t}} \operatorname{Ei} \left(-\frac{(b+x)^2}{4\kappa t} \right) - \frac{b-x}{2\sqrt{\pi \kappa t}} \operatorname{Ei} \left(-\frac{(b-x)^2}{4\kappa t} \right) \right\} \quad (6.14)$$

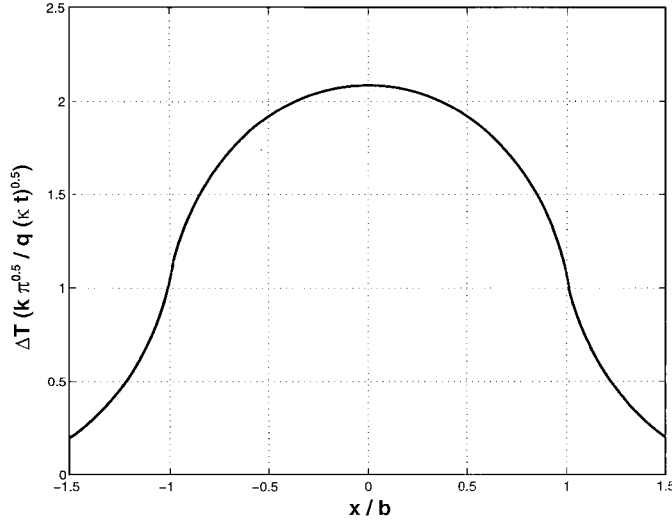


FIGURE 6.4 Temperature distribution across a heated strip of width $2b$ on the surface of a stationary semi-infinite solid. Curve is for case $\kappa t = b^2$.

where the error function

$$\text{erf}(X) = \frac{2}{\sqrt{\pi}} \int_0^X e^{-u^2} du$$

and the exponential integral

$$\text{Ei}(-X) = - \int_X^\infty \frac{e^{-u}}{u} du$$

A plot of Equation 6.15 is given in [Figure 6.4](#). It can be seen that the peak temperature rise occurs at the center of the strip ($x = 0$) and is given by

$$\Delta T_{\max} = \frac{2q}{k} \left(\frac{\kappa t}{\pi} \right)^{1/2} \left[\text{erf} \frac{b}{\sqrt{4\kappa t}} - \frac{b}{\sqrt{4\pi\kappa t}} \text{Ei} \left(-\frac{b^2}{4\kappa t} \right) \right] \quad (6.15)$$

At large times ($t \rightarrow \infty$) Equation 6.15 approaches a steady-state value

$$\Delta T_{\max_{ss}} = \frac{2qb}{k\sqrt{\pi}} \quad (6.16)$$

6.2.1.1.2.3 Constant Heat Supply q on Circular Region of Radius a on Surface $z = 0$ ([Figure 6.5](#))

In this case the steady-state temperature rise inside the heated circle on the surface $z = 0$ is given by:

$$\Delta T = \frac{2qa}{\pi k} \int_{\varphi=0}^{\pi/2} \sqrt{1 - \left(\frac{r}{a} \right)^2} \sin^2 \varphi d\varphi \quad (6.17)$$

where $x^2 + y^2 = r^2$ and $r \leq a$.

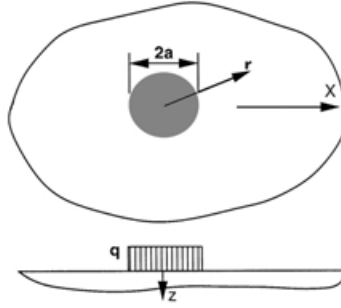


FIGURE 6.5 Circular heat source on surface of stationary body.

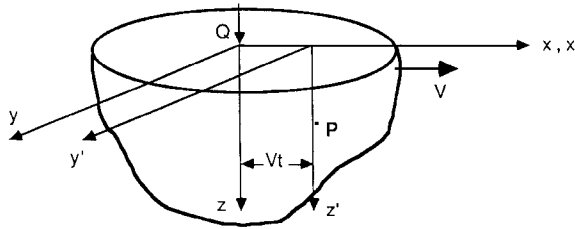


FIGURE 6.6 Point source of heat on the surface of a semi-infinite solid moving in x -direction with velocity V .

At the center of the circle, $r = 0$ and the temperature rise reaches a maximum

$$\Delta T_{\max} = \frac{qa}{k} \quad (6.18)$$

We note that for each of the cases of distributed heat flux, the temperature on the surface where heat is being applied remains finite.

6.2.1.2 Stationary Heat Source on Moving Body (or Moving Heat Source on Stationary Body)

This problem is treated by Carslaw and Jaeger (1959). The problem of a moving heat source on a stationary body is equivalent to the problem of a stationary heat source on a moving body. The important concept is that there is relative motion between the source of heat and the body into which the heat flows (convective diffusion).

6.2.1.2.1 Moving Semi-infinite Body with Stationary Continuous Point Heat Source (Figure 6.6)

Following the treatment of Carslaw and Jaeger (1959), define two coordinate systems as follows:

- Fixed x, y, z system, with origin at the stationary heat source
- Moving x', y', z' system, fixed in body moving with velocity V

The two coordinate systems are related by the expressions:

$$x' = x - Vt \quad y' = y \quad z' = z$$

Therefore, the temperature at a point P at time t is $T(x, y, z, t) = T(x' + Vt, y, z, t)$.

Fourier's law for heat conduction in the moving body is Equation 6.3, where now the material derivative is given by

$$\frac{DT}{Dt} = \frac{\partial T}{\partial t} + V \frac{\partial T}{\partial x}$$

Assume that the surface of the moving body is insulated except at the point heat source, which has strength Q . (**Note:** Gecim and Winer (1985) and others have found that convective cooling of the surface causes a negligible effect on flash temperature rise, so it is appropriate to assume an insulated surface when determining ΔT_f .)

The solution for this case is (Carslaw and Jaeger, 1959)

$$\Delta T = \frac{Q/\rho C}{\pi^{3/2} \kappa R} e^{\frac{Vx}{2\kappa}} \int_{\xi=R/\sqrt{\kappa t}}^{\infty} e^{-\xi^2 - V^2 R^2 / 16 \kappa^2 \xi^2} d\xi \quad (6.19)$$

where $R = x^2 + y^2 + z^2$

As time $t \rightarrow \infty$, the temperature rise ΔT approaches its quasi-steady-state value ΔT_{ss} .

$$\Delta T_{ss} = \frac{Q}{2\pi k R} e^{-V(R-x)/2\kappa} \quad (6.20)$$

As was the case with the stationary body, these solutions are not valid at the heat source ($R = 0$), which was assumed to act at an infinitesimally small point. A more realistic condition is the case of a distributed heat source, with the heat being distributed in some manner over a finite area. Any problem involving a distributed heat source on the surface of a moving semi-infinite body can be solved by integrating Equation 6.20 for the quasi-steady-state case or Equation 6.19 for the transient case.

6.2.1.2.2 Uniform Band Source Acting Over $-b \leq x \leq b$, $-\infty \leq y \leq \infty$ on Moving Body

Consider a semi-infinite body moving with velocity V in the x -direction, with a stationary heat source providing a flux q over the band (Figure 6.7). The quasi-steady-state temperature rise for this case is found by integrating the point source solution Equation 6.20. The result is (Carslaw and Jaeger, 1959):

$$\Delta T = \int_{-b}^b \frac{q}{\pi k} e^{\frac{V(x-x')}{2\kappa}} K_0 \left\{ \frac{V}{2\kappa} \left[(x-x')^2 + z^2 \right]^{1/2} \right\} dx' \quad (6.21)$$

where $K_0(\)$ is the modified Bessel function of the second kind, order 0. This result (Equation 6.21) is plotted in Figure 6.8. It can be seen that the results depend very much on the nondimensional Peclet number

$$Pe \equiv \frac{Vb}{2\kappa} \quad (6.22)$$

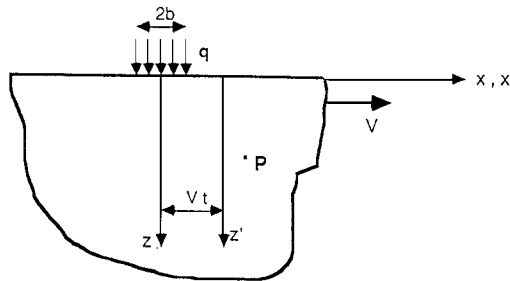


FIGURE 6.7 Uniform band heat supply on surface of semi-infinite solid moving in x -direction with velocity V .

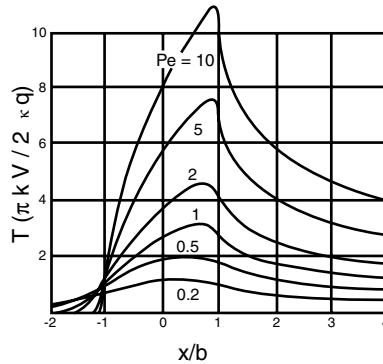


FIGURE 6.8 Surface temperature rise for a semi-infinite solid caused by friction at a contact of width $2b$ over which it slides with velocity V .

At large values of the Peclet number, say $Pe \geq 10$, the maximum temperature is seen to occur at the trailing edge. In such a case the maximum temperature rise at the trailing edge $x = b$ is given by

$$\Delta T_{\max} = \frac{2qb}{k\sqrt{\pi Pe}} \quad (6.23)$$

where $Pe \geq 10$.

It can be noted from Equation 6.23 that for constant q the maximum surface temperature decreases as the velocity increases (or Pe increases). This is due to the nature of heat transfer to a moving body. The material entering the heat source at the leading edge ($x = -b$) is at the relatively cool nominal surface temperature. That material has a finite heat capacity and thermal diffusivity, so a finite time is required to absorb the heat that causes an increase in its temperature. As the body's velocity increases, there is less time spent beneath the heat source by a given volume of material, and thus the temperature increase of the material will be smaller.

If the temperature is evaluated at different depths z , one would find that the temperature decreases very rapidly as $z > 0$, especially at high sliding velocities or high Pe . The heat that enters the moving body beneath the heat source is concentrated in a thin zone (the *thermal boundary layer*) under the heat source. This is shown in Figure 6.9.

For values of Peclet number less than 10, a closed-form expression for maximum temperature rise (equivalent to Equation 6.23) is not as easily found, but approximate expressions have been developed

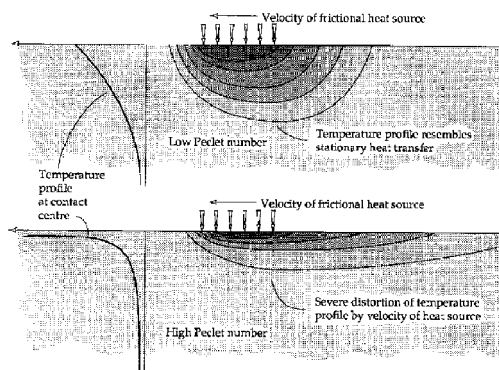


FIGURE 6.9 Temperature distribution in a moving body at low and high Peclet numbers. (From Stachowiak, G.W. and Batchelor, A.W. (1993), *Engineering Tribology*, Elsevier, Amsterdam. With permission.)

by Kuhlmann-Wilsdorf (1987), Greenwood (1991), and Tian and Kennedy (1994). For a uniform band source of heat on the surface of a body moving relative to the heat source with velocity V , the maximum flash temperature rise can be approximated as follows (Tian and Kennedy, 1994):

$$\Delta T_{\max} \approx \frac{2qb}{k\sqrt{\pi(1+Pe)}} \quad (6.24)$$

6.2.1.2.3 Moving Source with Uniform Heat Flux on Circular Region of Radius a

The solution for the case of uniform circular heat source on a moving body can be found using the results of Jaeger (1942). The maximum flash temperature rise for high Peclet number ($Pe > 10$) is:

$$\Delta T_{\max} \approx \frac{2qa}{k\sqrt{\pi Pe}} \quad (6.25)$$

where in this case the Peclet number is defined as:

$$Pe \equiv \frac{Va}{2\kappa} \quad (6.26)$$

For the entire range of Peclet number, the maximum steady-state flash temperature rise can be approximated as (Greenwood, 1991):

$$\Delta T_{\max} \approx \frac{2qa}{k\sqrt{\pi(1.273+Pe)}} \quad (6.27)$$

6.2.1.3 Summary of Solutions for Maximum Temperature Rise with Heat Sources of Various Shapes

The maximum flash temperature rise for a number of different contact shapes and pressure distributions has been determined by many investigators, including Blok (1937), Jaeger (1942), Archard (1958/59), Kuhlmann-Wilsdorf (1987), Greenwood (1991), Tian and Kennedy (1994), and Bos and Moes (1994). Generally they integrated expressions for the temperature rise due to a single heat source on a stationary or moving body (Equations 6.8 or 6.20, respectively) to obtain the maximum surface temperature within the heated region. A compilation of their solutions is given in [Table 6.1](#) for the entire range of Peclet number, which is defined by Equation 6.22 for the case of band or square heat source or by Equation 6.26 for circular or elliptical heat source ([Figure 6.10](#)). It should be noted that contacting asperities which are plastically deformed have a contact pressure distribution which is approximately uniform, giving a uniform heat flux, whereas elastic contacts have a Hertzian contact pressure distribution, which results in a parabolic or semi-ellipsoidal heat flux distribution.

Expressions for average temperature rise in the contacts can also be determined, and such expressions can be found in Archard (1958/59), Greenwood (1991), Tian and Kennedy (1994), and Bos and Moes (1994).

The expressions given in [Table 6.1](#) can be used in the determination of contact temperature by the methodology to be described below.

6.2.1.4 Flash Temperature Transients

Although methodology is presented above for determining either transient or steady-state flash temperature rises, the expressions in [Table 6.1](#) are for steady or quasi-steady-state conditions. Since the initial work by Blok (1937) and Jaeger (1942), nearly all investigators have assumed that steady conditions prevail. There are two reasons for using the steady-state or quasi-steady-state assumption:

TABLE 6.1 Expressions for Maximum Flash Temperature Rise for Various Heat Source Distributions

Shape of Heat Source	Heat Flux Distribution	Figure No.	Maximum Flash Temperature Rise (Steady State)		
			Stationary or Low Speed $Pe < 0.1$	High Speed $Pe > 10$	Approximate Expression for All Velocities
Band	Uniform	6.7	$\frac{2qb}{k\sqrt{\pi}}$	$\frac{2qb}{k\sqrt{\pi Pe}}$	$\frac{2qb}{k\sqrt{\pi(1+Pe)}}$
Square	Uniform	6.10a	$\frac{1.122qb}{k}$	$\frac{2qb}{k\sqrt{\pi Pe}}$	$\frac{2qb}{k\sqrt{\pi(1.011+Pe)}}$
Circular	Uniform	6.5	$\frac{qa}{k}$	$\frac{2qa}{k\sqrt{\pi Pe}}$	$\frac{2qa}{k\sqrt{\pi(1.273+Pe)}}$
Circular	Parabolic	6.10b	$\frac{3\pi\bar{q}a}{8k}$	$\frac{2.32\bar{q}a}{k\sqrt{\pi Pe}}$	$\frac{2.32\bar{q}a}{k\sqrt{\pi(1.234+Pe)}}$
Elliptical	Uniform	6.10c	$\frac{qa}{k\sqrt{Se}}$	$\frac{2qa}{k\sqrt{\pi Pe}}$	$\frac{2qa}{k\sqrt{\pi(1.273Se+Pe)}}$
Elliptical	Semi-ellipsoidal	6.10d	$\frac{3\pi\bar{q}a}{8k\sqrt{Se}}$	$\frac{2.32\bar{q}a}{k\sqrt{\pi Pe}}$	$\frac{2.32\bar{q}a}{k\sqrt{\pi(1.234Se+Pe)}}$

Pe is the Peclet number given by Equation 6.22 for band or square contacts and by 6.26 for circular or elliptical contacts, $\bar{q}$ is the mean heat flux, Se is a shape function for elliptical heat sources, given by

$$Se = \frac{16e^{1.75}}{(3 + e^{0.75})(1 + 3e^{0.75})}$$

and $e = b/a$ is the aspect ratio of the elliptical source.

Source: Adapted from Jaeger, 1942; Kuhlmann-Wilsdorf, 1987; Greenwood, 1991; Tian and Kennedy, 1994; and Bos and Moes, 1994.

1. The steady-state temperature is the largest flash temperature rise that can occur, so that assumption will result in conservative estimates of maximum surface temperatures.
2. The steady-state condition is reached in a very short time after sliding commences, so nearly all of the time of sliding is spent in steady-state conditions.

It is of interest to confirm the validity of the second assumption and to know how long it takes for steady-state conditions to be reached. Jaeger (1942) evaluated the transient temperature at the center of a square heat source moving at a velocity V over a surface, and the results are shown in Figure 6.11. It can be seen that the temperature rapidly approaches the steady-state value at a rate that is dependent on Peclet number. Bhushan (1987a) analyzed those results and concluded that the flash temperature reaches its steady-state value after moving a distance of only 1.25 times the length of the heat source. More recently, Yevtushenko et al. (1997) studied the transient temperature rise for a moving circular heat source and concluded that the temperature reaches at least 87% of its steady-state value almost instantaneously (within approximately one heat source length for a Peclet number of 1) and then asymptotically approaches the steady state. The duration required to reach steady-state conditions decreases as the sliding velocity (or Peclet number) increases; the longest duration is that for a stationary heat source. Gecim and Winer (1985) analyzed the case of a stationary circular source and concluded that the surface temperature is a function of the Fourier number, $F = \kappa t/R^2$. The maximum temperature (at the center

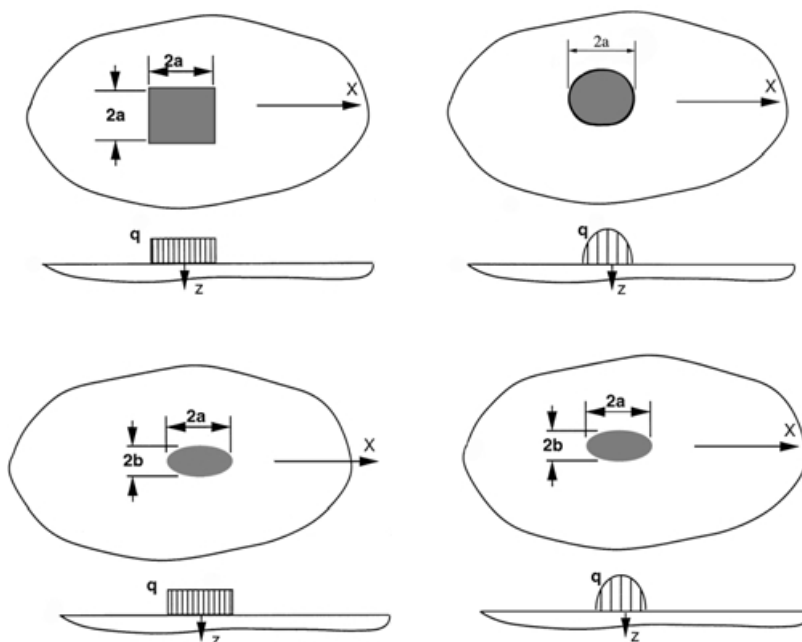


FIGURE 6.10 Diagrams of heat sources used in Table 6.1. (a) Square heat source with uniform heat flux distribution. (b) Circular heat source with parabolic heat flux distribution. (c) Elliptical heat source with uniform heat flux distribution. (d) Elliptical heat source with semi-ellipsoidal heat flux distribution.

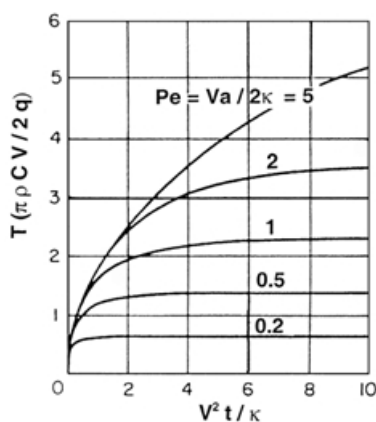


FIGURE 6.11 Surface temperature at center of a moving square heat source as function of time and Peclet number. (Adapted from Jaeger, J.C. (1942), Moving sources of heat and the temperature of sliding contacts, *Proc. R. Soc. NSW*, 76, 203-224.)

of the source) reaches about 95% of its steady-state value before $F = 25$ and then gradually approaches the steady value by the time $F = 100$ (Gecim and Winer, 1985).

Based on these considerations, in nearly all frictional heating situations it can be safely assumed that the steady-state (or quasi-steady-state) conditions prevail, and the solutions listed in Table 6.1 can be used. In cases involving contacts of very short duration, however, it may be necessary to use transient solutions of the heat source equations. For example, Hou and Komanduri (1998) found that in three-body abrasive finishing operations the contact times between an abrasive grit and a surface being polished or ground are so short that transient surface temperature solutions may be required.

6.2.1.5 Nominal Surface Temperature Rise

If a single concentrated heat source does not repeat the same path over the contact surface, and if the body is very large, there is no significant nominal surface temperature rise because all heat generated at the contact spot is transferred into the three-dimensional heat sink. The contact temperature in that case can be found, without significant error, by adding the local (flash) temperature rise to the bulk temperature of the body, as was done in Blok's (1937) analysis. However, if the heat source passes repeatedly over the same point on the surface, there unavoidably exists an extra surface temperature rise because the frictional heat generated during one pass cannot all flow away from the contact area before the next generation of heat at the same spot. The extra surface temperature rise is the nominal surface temperature rise. (Some others, such as Ashby, et al. (1991), have referred to this as a bulk surface temperature rise.)

In determining the nominal surface temperature rise, an imaginary heat flux could be considered to be evenly distributed across the entire nominal contact area on the stationary body and another imaginary heat flux applied to the contacting surface of the moving body (Tian and Kennedy, 1993). The thermal response of the bodies to those heat fluxes would then be analyzed. For the case of a fast-moving heat source which repeatedly sweeps over the same path (as for a pin on a wear track on a disk or a brake pad on a rotor), or for a heat source which oscillates over the same contact path, the heat flux on the moving body should be applied to the entire area swept out by the contact. For a slow speed case, the imaginary heat flux would be applied only to the nominal contact area. While surface heat convection has little influence on the local temperature rise, the nominal temperature rise is affected by large-scale boundary conditions such as convection or conduction, as well as the total heat entering the body. Therefore, the nominal surface temperature rise on one of the contacting bodies can be expressed as:

$$\Delta T_{nom} = f(Q, A_{nom}, h, t, G) \quad (6.28)$$

where Q is the total heat entering the contact surface of the body due to friction and other heat sources, A_{nom} is the nominal contact area, h is heat convection coefficient at boundaries, t is the time during which the frictional heat is applied, and G is a geometrical and material factor related to the shape of the finite body and thermal properties of the material.

In most cases the nominal temperature rise can be determined by an approximate one-dimensional analysis in which the heat flux is spread over the nominal contact area A_{nom} if the body is not moving relative to the heat source. If the body is moving relative to the heat source, A_{nom} is replaced by A_{swept} which is the entire area swept out by the heat source.

This concept is best illustrated by examples. Consider the case of a stationary pin of circular cross-section sliding against a rotating disk (Figure 6.12). The pin, which is stationary with respect to the contact zone, will be considered first. It is assumed to be held in place by a large heat sink at bulk temperature T_{b1} and has heat flux q_1 entering the real area of contact A_r on its surface. The nominal area

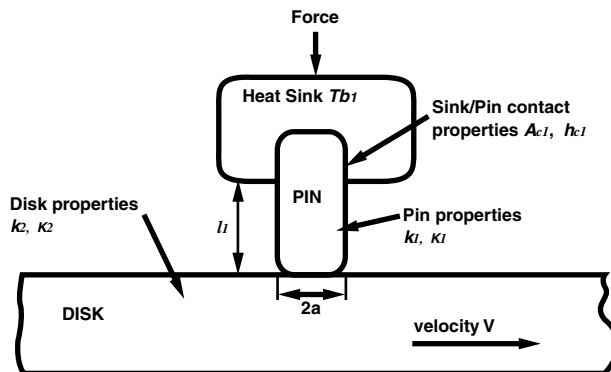


FIGURE 6.12 Schematic diagram of a typical pin-on-disk contact.

of contact is $A_{nom1} = \pi a^2$. If the nominal area of contact is larger than the real area of asperity contact, the nominal heat flux into the stationary pin is:

$$q_{nom1} = q_1 A_r / A_{nom} \quad (6.29)$$

Assuming that the pin has been in contact long enough to come to thermal steady state, then the heat flow in the pin is approximately linear and the nominal surface temperature rise for that body is given by (Ashby et al., 1991):

$$\Delta T_{nom1} = T_{nom1} - T_{b1} = q_{nom1} l_{b1} / k_1 \quad (6.30)$$

If there is very good thermal contact between pin and heat sink, the heat conduction length l_{b1} is equal to the pin length l_1 . Imperfect thermal contact between pin and heat sink can lead to higher nominal temperature rises for the stationary body. This can be taken into account by adjusting the heat conduction length using the method of Ashby et al. (1991). If the effective heat transfer coefficient between pin and clamp is h_{c1} (units: W/m²K), then simple continuity of heat fluxes gives:

$$l_{b1} = l_1 + \frac{A_n k_1}{A_{c1} h_{c1}} \quad (6.31)$$

where A_{c1} is the nominal area of the clamp contact. Ashby et al. (1991) found that the effective heat conduction length l_{b1} can often be double the actual length l_1 .

The nominal heat flow into the moving disk can be determined in several ways. Ashby et al. (1991) consider that the heat source is injecting heat into a point on the disk surface for only a transit time, after which the heat diffuses into the bulk of the body before the same point is heated again on the next traversal past the heat source. They determine an effective diffusion distance l_{b2} , which is given by the following expression (Ashby et al., 1991):

$$l_{b2} = \frac{a}{\pi^{1/2}} \tan^{-1} \left[\frac{2\pi\kappa_2}{aV} \right]^{1/2} \quad (6.32)$$

Then the nominal surface temperature rise for the moving body is determined by:

$$\Delta T_{nom2} = T_{nom2} - T_{b2} = q_{nom2} l_{b2} / k_2 \quad (6.33)$$

An alternative approach which is particularly useful for relatively rapidly moving bodies was suggested by Tian and Kennedy (1993). The nominal heat flux for a moving body can be modeled as an imaginary, evenly distributed, stationary heat source over the whole sliding surface (A_{swept}). The imaginary average stationary heat flux in this case equals (from Equation 6.29):

$$q_{nom2} = q_2 A_r / A_{swept} \quad (6.34)$$

where A_{swept} is the total surface area swept out by the nominal sliding track on the contact surface of the moving disk. For example, if the wear track on a disk surface has a mean radius r_m (Figure 6.13) then the swept area is $A_{swept} = 2\pi r_m (d_n)$, where d_n is the nominal width of the contact in the direction perpendicular to sliding. The solution for the nominal surface temperature using the imaginary heat flux Equation 6.24 can usually be found by solving a one-dimensional heat conduction problem for the thickness of the disk, resulting in Equation 6.33.

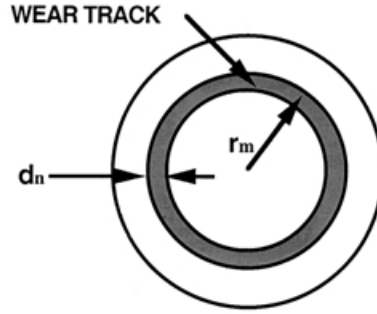


FIGURE 6.13 View of normal sliding track (wear track) on moving disk surface.

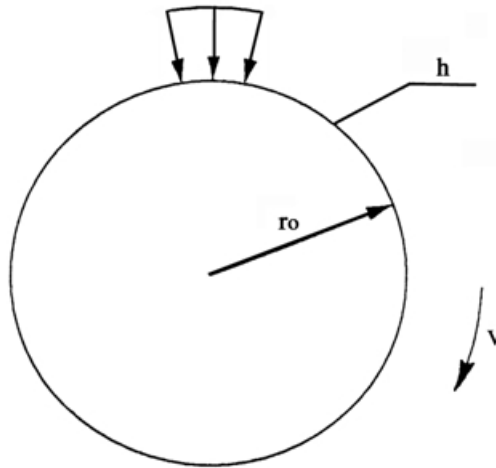


FIGURE 6.14 Stationary band heat source on outside surface of rotating cylinder.

If convection occurs over the surface of the moving disk, and if the average convection coefficient over the whole contact surface is h , it can be shown that the steady-state temperature rise above the background temperature is (Tian and Kennedy, 1993):

$$\Delta T_{nom2} = T_{nom2} - T_{b2} = q_{nom2} / h \quad (6.35)$$

A similar equation results for the case of a stationary heat source acting on the surface of a fast-rotating cylinder as shown in Figure 6.14. Assume that the nominal width of the heat source is equal to the width of the cylinder and that there is no heat loss from the circular ends of the cylinder. The average convection coefficient over the contacting surface is h . Using the present surface temperature model, the swept area on the cylinder surface is $A_{swept} = 2\pi r_o (w)$, where r_o is the mean radius of the cylinder and w is the axial width of the nominal contact area (and of the cylinder). Equations 6.34 and 6.35 are still valid for determining the nominal heat flux and nominal surface temperature rise, respectively, for this case (Tian and Kennedy, 1993).

6.2.1.6 Partition of Frictional Heat

When frictional heating occurs at a contact interface during sliding, it is necessary to determine the partition of frictional heat between the two contacting bodies in order to predict the surface temperature rise using the aforementioned solutions for temperature in a single body.

We will assume that all of the frictional heat, given by Equation 6.2, is generated at the contact interface within the real area of contact. Some of the generated heat goes into Body 1, which is moving at velocity V_1 relative to the contact area, and some into Body 2, which is moving at velocity V_2 relative to the contact.

Define

q_1 = heat flux (heat per unit time per unit area) into Body 1

q_2 = heat flux into Body 2

Then

$$q_1 + q_2 = q_{total} = \mu p U \quad (6.36)$$

where $U = |V_2 - V_1|$.

Let us define heat partitioning factor α :

$$\alpha = q_1 / q_{total} \quad \text{and} \quad 1 - \alpha = q_2 / q_{total} \quad (6.37)$$

Then

$$q_1 = \alpha \mu p U \quad \text{and} \quad q_2 = (1 - \alpha) \mu p U \quad (6.38)$$

In general, α is a function of position ($\alpha = \alpha(x, y)$) and both q_1 and q_2 can vary with position. In fact, q_{total} may also vary within the contact because the contact pressure is not necessarily constant and because the sliding velocity may vary with position.

6.2.1.6.1 Complete Solutions for Heat Partition Function

If contact between the two bodies is perfect within the real contact area, no temperature jump should be expected across the contact interface within the real area of contact. The most accurate heat partition function $\alpha(x, y)$ can be obtained by matching the surface temperatures of the two contacting solids at all points within the real contact area. This has been done by several investigators, particularly Ling (1959) and Bos and Moes (1995). Ling's methodology involved an iterative solution in which transform solutions are used for the temperatures on each surface (see Section 6.2.2.1 below) and then the contact temperature is matched at each point on the two surfaces (Ling, 1973). Bos and Moes (1995) developed a numerical algorithm to determine the heat partitioning function for arbitrarily shaped contact by matching the surface temperatures of the two contacting bodies at all points within the contact area. They used heat source methods for determining the temperature solutions for each surface.

An example of the result of a temperature-matching algorithm is shown in Figure 6.15. The dashed lines show the surface temperature rise for a square heat source on moving and stationary bodies, as determined using expressions given in Table 6.1. Those results have been normalized to give the same maximum temperature rise. The solid curve shows the temperature distribution for the same conditions using Ling's iterative procedure. It is apparent that the actual temperature distribution is between those predicted for the stationary and moving surfaces. Most of the frictional heat dissipated at the leading edge of the contact flows into the moving body, because the material of that body is cool when it enters the contact and it needs to be brought quickly up to the temperature of the stationary surface with which it is in contact. On the other hand, by the time the moving surface reaches the trailing edge of the contact, much less of the frictional heat enters the already heated moving surface.

6.2.1.6.2 Approximate Solutions for Heat Partition Using the Blok Postulate

Owing to the difficulty of determining the complete heat partitioning function $\alpha(x, y)$, most investigators have made use of an approximation first made by Blok (1937), in which it is assumed that the heat partitioning function is a constant factor. Blok postulated that overall heat partitioning factor can be

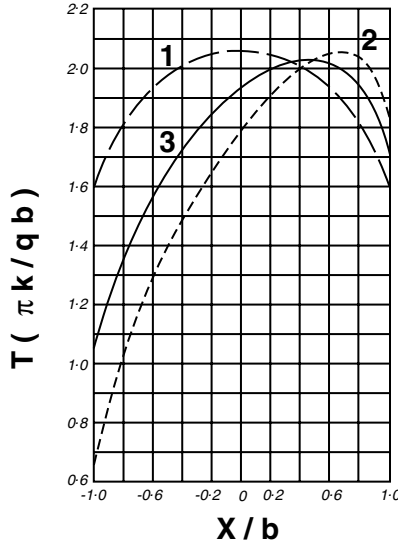


FIGURE 6.15 Temperature distributions resulting from square heat source at interface between moving and stationary bodies. Curve 1: stationary body solution; Curve 2: moving body; Curve 3: iterative solution with temperature matching everywhere in contact. (From Ling, F.F. (1973), *Surface Mechanics*, John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.)

estimated by setting the maximum surface temperature of the two bodies equal within the contact (Blok, 1937; Ling, 1973). Various analyses have been carried out by others to determine the exact heat partitioning function; it was found that the computed values of partitioning factor were in good agreement with the results obtained using Blok's postulate (Ling, 1973; Bos and Moes, 1995).

6.2.1.6.2.1 Case of Band Contact (Two-Dimensional Heat Flow)

Consider two large bodies, one of which is stationary ($V_1 = 0$), the other of which is moving with velocity U ($V_2 = U$). The bodies are in contact along the band $-\infty \leq y \leq \infty$ and $-b \leq x \leq b$, and the two bodies have the same sum of bulk temperature and nominal surface temperature rise (that sum could be equal to 0). The temperature on the surface of the stationary body is given by Equation 6.14 and Figure 6.4. For the moving body the surface temperature is given by Equation 6.21 and Figure 6.8. By comparing the temperature distributions in Figures 6.4 and 6.8 one can see that they are quite different in shape, so equating the temperatures at all points within the region $-b \leq x \leq b$ is not possible using those distributions. Instead, we will make use of the Blok postulate and set T_{max} equal on moving and stationary surfaces. For the band contact case, Equations 6.16 and 6.24 can be used and this gives:

for the stationary body

$$T_{max} = \frac{2q_1 b}{k_1 \sqrt{\pi}} \quad (6.16a)$$

and for the moving body

$$T_{max} = \frac{2q_2 b}{k_2 \sqrt{\pi(1+Pe)}} \quad (6.24a)$$

where

$$Pe = \frac{Ub}{2\kappa_2}$$

Equating these gives

$$\alpha = \frac{1}{1 + \frac{k_2}{k_1} \sqrt{1 + Pe}} \quad (6.39)$$

As can be seen, for high sliding velocities Pe is large and α is small. Thus, at high sliding velocities most of the heat enters the moving body (the body moving with respect to the heat source). This is because the moving body presents more new material per unit time to the heat source, where it must then be heated up to T_{max} . The surface of the stationary body is always at temperature T_{max} and need not receive much heat from the source to remain at that temperature.

Using the Blok postulate, then

$$q_1 = \frac{\mu p U}{1 + \frac{k_2}{k_1} \sqrt{1 + Pe}} \quad (6.40)$$

and this can be used in Equation 6.16a to find the surface temperature.

It might be noted that if sliding speeds are very low ($Pe < 0.1$), then at each position within the contact there is ample time for the temperature distribution in the moving body to approach that of the stationary body. In that case, one finds

$$\alpha = \frac{1}{1 + \frac{k_2}{k_1}} \quad (6.41)$$

So, at very low sliding speeds the conductivities of the sliding bodies govern the partitioning of the frictional heat. The majority of the frictional heat enters the body with the highest thermal conductivity.

6.2.1.6.2.2 Case of Circular Contact Spots (Three-Dimensional Heat Flow)

In most actual sliding situations, the real area of contact is composed of a number of small circular or elliptical junctions. If one assumes that the contact spots are well separated and their temperature fields do not interact, the contact temperature solution requires use of an expression such as Equation 6.27 for both bodies. As was shown above for the band contact case, the two expressions could then be equated using the Blok postulate, as long as there is no nominal temperature rise for either body ($\Delta T_{nom2} = \Delta T_{nom1} = 0$) and both bodies have the same bulk temperature ($T_{b1} = T_{b2}$).

Let us consider the case of contacting hemispherical asperities, both of which are moving with respect to the circular contact area where heat is being generated. The Peclet number Pe_i of each of the two bodies is found by using its relative velocity V_i in Equation 6.26. The contact radius is a .

Using the solution for the case of a uniform circular heat source of radius a (as in Equation 6.27) for each body, the maximum interface temperature is given by

$$T_{c\max} = \frac{2a\mu p U}{\sqrt{\pi} \left[k_1 \sqrt{(1.273 + Pe_1)} + k_2 \sqrt{(1.273 + Pe_2)} \right]} \quad (6.42)$$

This is the case encountered when there is contact between hemispherical asperities of two materials and there is plastic deformation of the softer of the two contacting materials. In this case, p = the indentation hardness of the softer material.

For elastic contact between hemispherical asperities, there is a Hertzian contact pressure distribution, and the maximum interface temperature can be derived using the solution for the case of a parabolic heat source, [Table 6.1](#).

$$T_{c\max} = \frac{1.31a\mu\bar{p}U}{\sqrt{\pi} \left[k_1 \sqrt{(1.2344 + Pe_1)} + k_2 \sqrt{(1.2344 + Pe_2)} \right]} \quad (6.43)$$

where $\bar{p}$ is the mean pressure of elastic contact.

6.2.1.6.3 General Contact Case

In general, the maximum contact temperature for a surface can be found using Equation 6.1. Using the Blok postulate, the maximum temperature is set equal for the two surfaces, giving:

$$T_{c\max 1} = T_{b1} + \Delta T_{nom1} + \Delta T_{f\max 1} = T_{b2} + \Delta T_{nom2} + \Delta T_{f\max 2} = T_{c\max 2} \quad (6.44)$$

The nominal temperature rise and maximum flash temperature rise for Body i can be found as a linear function of an unknown heat flux q_i entering that surface, using the methods of Section 6.2.1.5 for ΔT_{nom} and [Table 6.1](#) for $\Delta T_{f\max}$. This will give expressions of the form $\Delta T_{nom1} = A_1 q_1$, $\Delta T_{nom2} = A_2 q_2$, $\Delta T_{f\max 1} = B_1 q_1$ and $\Delta T_{f\max 2} = B_2 q_2$, where A_i and B_i are influence coefficients which depend on contact geometry, sliding velocity and thermal properties. For example, if the contact is circular and the pressure distribution is uniform, from [Table 6.1](#):

$$B_1 = \frac{2a}{k_1 \sqrt{\pi(1.273 + Pe_1)}} \quad \text{and} \quad B_2 = \frac{2a}{k_2 \sqrt{\pi(1.273 + Pe_2)}}$$

Using the A_i and B_i in Equation 6.44, one gets

$$T_{c\max} = T_{b1} + A_1 q_1 + B_1 q_1 = T_{b2} + A_2 q_2 + B_2 q_2 \quad (6.45)$$

Defining the heat partition factor α as above, expressions 6.38 can be used to get 6.45 in terms of α , and the total heat flux rate, $q_{total} = \mu p U$. Solving for α , one gets:

$$\alpha = \frac{(T_{b2} - T_{b1}) + q_{total}(A_2 + B_2)}{q_{total}(A_1 + A_2 + B_1 + B_2)} \quad (6.46)$$

It can be seen easily that Equation 6.46 reduces to 6.39 for the case of band contact with $T_{b1} = T_{b2}$ and $\Delta T_{nom1} = \Delta T_{nom2} = 0$ (so that $A_1 = A_2 = 0$).

Therefore, for the general contact case, one can determine the heat partition factor using Equation 6.46 and then use it, A_1 and B_1 , along with the heat flux $q_1 = \alpha q_{total} = \alpha \mu p U$, in Equation 6.45 to find the maximum contact temperature.

6.2.2 Numerical Methods for Surface Temperature Determination

Several important assumptions were implicit in the development of the heat source equations in Section 6.2.1 above. Of particular importance are the assumptions that the body is very large and is

homogeneous (since the heat source solutions are for a homogeneous, semi-infinite, half-space. Real contacting bodies may be large relative to the contact size, but they are not infinitely large, and they often contain more than one material. To overcome that difficulty, numerical methods have been developed to solve the heat conduction equation (6.3) for bodies of finite dimensions involving frictional heating over a portion of their surface. These include integral transform methods, finite element methods, and hybrid methods which combine both integral transforms and finite elements. When used appropriately, these methods allow natural temperature matching on the two contacting surfaces without resorting to artificial assumptions concerning heat partitioning. They also allow the determination of contact temperatures in layered bodies and determination of temperature distribution throughout the contacting solids. This makes them a valuable companion to numerical stress analysis programs for the determination of thermomechanical effects (deformation and stress distributions) for sliding systems subjected to frictional heating. This generally is not possible (or is inconvenient) with the heat source methods described above.

6.2.2.1 Integral Transform Methods

Integral transform methods for surface temperature determination were first developed by Ling and co-workers (Ling, 1973), and were later applied by Floquet et al. (1977) and others. Of particular importance in sliding contact problems are Fourier transform-based methods. The method enables the solution of the heat conduction equation (6.5) in each of the two contacting bodies, subject to appropriate boundary conditions. Temperatures are matched at the contact interface between the bodies, making it unnecessary to specify the heat partition function, and the total heat flux distribution, q_{total} , is applied at the contact interface. A transform which is appropriate to the geometry of the problem is applied to the equations and boundary conditions, the transformed equations are solved, and then the solution is inverted to obtain the temperature distribution.

Transform methods have been found to give good results for contact temperatures and temperature distributions for the complete range of velocities which are encountered in tribological applications of real solid bodies.

An example of the use of integral transform techniques is shown in [Figure 6.16](#) (Floquet et al., 1977). Both shaft and housing are cylindrical, and different types of motion were prescribed for the system. The results point out that surface temperatures are greater if the more conductive component (the shaft) is stationary than if the same component is moving relative to the contact zone. Integral transform methods have proven very useful and accurate for determining contact temperatures in cylindrical bearings in both two dimensions (Floquet et al., 1977) and three dimensions (Floquet and Play, 1981). Other examples of integral transform applications are given by Ling (1973). Several investigators have combined integral transform solutions for temperature distributions with transform solutions of the elasticity equations to determine thermoelastic contact stress distributions for various contact problems (e.g., Ling, 1973; Ju and Huang, 1982).

Although transform techniques do not generally require large computational times or storage requirements, there is no readily available transform-based software for performing the analysis, so considerable time is required to generate the computer code. Another major disadvantage is that the integral transform techniques are only applicable to simple geometries (cylindrical shaft and housing, rectangular bodies, etc.), and this limits their usefulness.

6.2.2.2 Finite Element Methods

Many commercial finite element codes enable solution of heat conduction problems. They can be used to solve Equation 6.3, but in just about all cases they assume that the body is not moving relative to the heat source ($V = 0$ in Equation 6.3). They can still be used for surface temperature analysis, but only if a transient problem is solved, with the heat source being considered fixed for a given time increment and moved for the next increment. A better way is to use a finite element program specifically written to solve the quasi-steady heat conduction equations (Equation 6.4 with $\partial T/\partial t = 0$). One such program is described by Kennedy (1981).

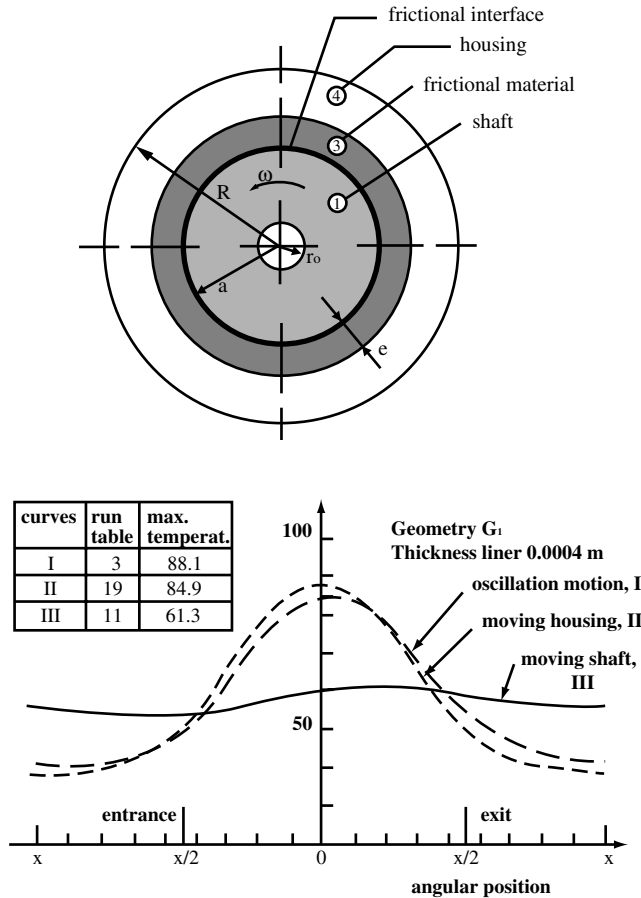


FIGURE 6.16 (a) Geometry used in transform-based analysis of surface temperatures in dry bearing. (b) Results of analysis for different types of motion (contact temperature vs. angular position). (From Floquet, A., Play, D., and Godet, M. (1977), Surface temperatures in distributed contacts: application to bearing design, *ASME J. Lubrication Technology*, 99, 277-283. With permission.)

The essence of the finite element method for surface temperature determination is as follows: finite element meshes are defined for the two contacting bodies. Those meshes share nodes within the real area of contact, and those nodes (contact nodes) enable temperature matching for the contacting surfaces throughout the contact region. The finite element meshes should be very fine in the contact region, where temperature gradients are highest. An example of such a mesh is shown in Figure 6.17. The desired distribution of frictional heat flux q_{total} is prescribed at the contact interface. That heat flux flows naturally into the two contacting bodies, so no heat partitioning function needs to be prescribed. The velocity of each body is used in the finite element version of Equation 6.5 to get the finite element equations for each element of that body. The finite element equations for the entire system are solved, using the appropriate boundary conditions, to find the temperature distributions in both bodies.

An example of a temperature distribution determined by finite element methods is shown in Figure 6.18. The particular case being analyzed is a ceramic-coated metallic face seal ring in contact with a carbon-graphite seal ring. The contact is assumed to be moving with the rotating metallic ring, so it is moving with respect to the stationary carbon-graphite ring (shown on the bottom in Figure 6.18). The temperature distribution in that case was used as input for a finite element stress analysis (thermo-elasto-plastic) program to study the thermal stresses that develop in the vicinity of a frictionally heated contact.

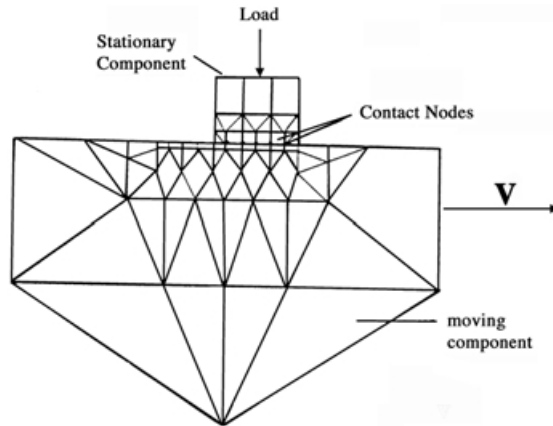


FIGURE 6.17 Typical finite element mesh for surface temperature analysis.

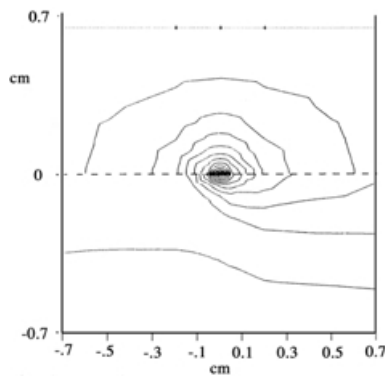


FIGURE 6.18 Finite element results for temperature distribution (isotherms) ceramic-coated metallic seal ring (top) in contact with carbon graphite seal ring (bottom). Circumferential direction is horizontal in figure; axial direction is vertical. Contact patch extends from -0.05 to $+0.05$ cm. Ceramic coating is 0.02 cm thick. (From Kennedy, F.E. and Hussaini, S.Z. (1987), Thermo-mechanical analysis of dry sliding systems, *Computers and Structures*, 26, 345-355. With permission from Elsevier Science.)

Finite element methods have proven very useful in a variety of frictional heating calculations, and they have many advantages for such calculations. They can be used with bodies of finite dimensions, and of real irregular geometries; they do not require artificial heat partitioning assumptions, and guarantee temperature matching throughout the contact region. Their output can be input easily to stress analysis programs, so they can be quite suitable for use in combined temperature-stress models.

Finite elements are not problem-free for surface temperature determination. In particular, at high sliding speeds, or high Peclet number, the results are subject to numerical instability (Kennedy et al., 1984). Those instabilities are caused by the convective-diffusion term ($\mathbf{V} \cdot \nabla T$) in Equation 6.4. Methods have been developed to help minimize those problems, but difficulties remain at high Peclet numbers (Kennedy et al., 1984).

6.2.2.3 Hybrid Methods

Integral transform methods are very useful for determining contact temperature distributions in simple geometries for any sliding velocities, but many tribological components, such as bearing housings, have relatively complex geometries which cannot be studied easily using integral transforms. On the other hand, finite element methods can easily be used for complex geometries but have problems with bodies moving at high sliding velocities. A hybrid technique which can address these difficulties was developed

by Colin and Floquet (1986), who took advantage of the fact that most moving tribological components, such as a rotating journal, have a relatively simple shape, while those which are stationary, such as a bearing housing, tend to be more complex in shape. The hybrid technique uses transform methods for the moving component and finite element methods for the stationary component.

Both finite element and integral transform methods give a relation between heat input and temperature. In the hybrid method, the stationary body is discretized with a finite element mesh and the moving body with a surface mesh whose nodes correspond with nodes of the finite element mesh within the contact zone. Similar interpolation functions are prescribed for temperature of the two surfaces within the contact zone, and the total heat flux distribution is prescribed on the contact interface. The methodology is described in detail by Colin and Floquet (1986).

An example of the application of the hybrid technique is shown in [Figure 6.19](#). It is seen that even with a coarse finite element mesh, it is possible to get valid results for a wide range of sliding velocities. It should be noted, however, that there is no readily available software for use in determining sliding contact temperatures using the hybrid method.

6.3 Surface Temperature Measurement

Although surface temperature analysis methods, both analytical and numerical, are well developed and widely used, they all suffer from a major difficulty: their use requires knowledge of the area within which the frictional heat is being generated. Seldom is that real area of contact known with any certainty *a priori*. As a result, it is often necessary to measure surface temperatures experimentally. Surface temperature measurement techniques could be grouped under three categories: point temperature sensors (e.g., thermocouples and thermistors), field radiation-based sensors (e.g., infrared sensors), and *ex post facto* observations. The various temperature measurement methods were described in several recent publications (Kennedy, 1992; Bhushan, 1999). A brief description of the techniques follows.

6.3.1 Thermocouples, Thermistors, and Related Temperature Sensors

6.3.1.1 Embedded Subsurface Thermocouples

Thermocouples are probably the most commonly used sensors for measuring temperatures, including those resulting from frictional heating. Their operation is based on the findings of Seebeck, who demonstrated in 1821 that a specific electromotive force (emf) potential exists as a property intrinsic to the composition of a wire, the ends of which are kept at two different temperatures. The simplest measuring circuitry for thermocouple thermometry involves wires of two dissimilar metals connected together so as to give rise to a total relative Seebeck potential. This emf is a function of the composition of each wire and the temperatures at each of the two junctions. This circuit can be well characterized such that, if one junction is held at a known reference temperature, the temperature of the other “measuring” junction can be inferred by comparison of the measured total emf with an empirically derived calibration table (Reed, 1982).

Generally, when the thermocouple technique is to be used to measure contact temperature, a small hole is drilled into a noncontacting surface, usually the rear side, of the stationary component of a frictional pair. The hole may extend to, or just beneath, the sliding surface. An adhesive is put in the hole, either ceramic cement for components encountering high sliding temperatures or a polymeric adhesive such as epoxy for lower temperature components. Ideally, the adhesive should be an electrical insulator but a reasonably good thermal conductor. A small thermocouple is then inserted in the hole so that its measuring junction rests either at or just beneath the sliding surface and is held in position by the cement, which also serves to insulate the thermocouple wires from the surrounding material. A diagram of a typical embedded thermocouple installation is shown in [Figure 6.20](#). Several such thermocouples can be embedded at different depths and at various locations along the sliding path to get information about surface temperature distribution and temperature gradients. By monitoring the thermocouple(s) throughout a sliding interaction, the transient nature of the surface temperatures can be

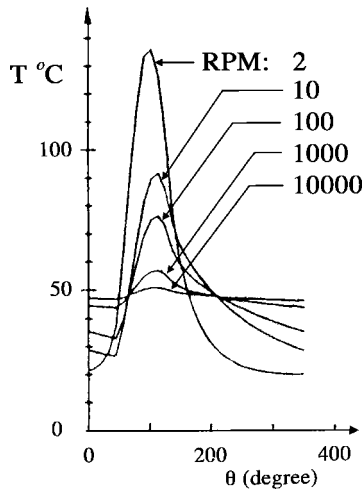
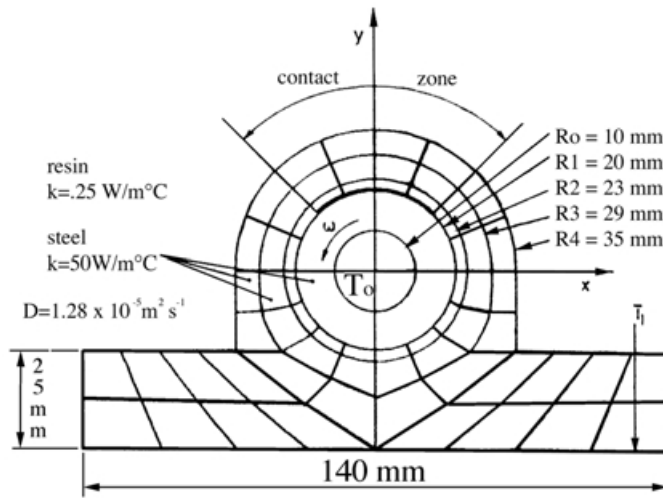


FIGURE 6.19 Hybrid numerical analysis of temperature distribution in bearing. Finite element mesh was used for bearing support, Fourier transforms for rotating shaft. Model configuration is at left. Resulting temperatures, as function of velocity, are at right. (From Colin, F. and Floquet, A. (1986), Combination of finite element and integral transform techniques in a heat conduction quasi-static problem, *Int. J. Numer. Methods in Engg.*, 23, 13-26. Reproduced by permission of John Wiley & Sons, Limited.)

deduced. It should be mentioned that subsurface thermocouples can also be embedded in the moving component of a friction couple, but slip rings or a similar means will be required to gain access to the thermocouple output.

Embedded thermocouples have been found to give a good indication of the transient changes in frictional heat generation which accompany gross changes in contact area (Ling and Simkins, 1963; Santini and Kennedy, 1975). They cannot, however, give a true indication of surface temperature peaks. Subsurface thermocouples have a limited ability to respond to flash temperatures owing to their finite thermal mass and distance from the points of intimate contact where heat is being generated. A thermocouple can be made part of the sliding surface by placing it in a hole which extends to the surface and then grinding the thermocouple flush with the surface. Even in that case, however, the finite mass of the thermocouple junction prevents it from responding to very short-duration flash temperature pulses

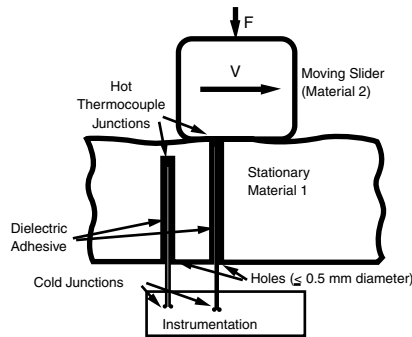


FIGURE 6.20 Typical installation of embedded thermocouples for measuring surface and near-surface temperatures resulting from frictional heating.

(Suzuki and Kennedy, 1988). In addition, the thermocouple and the hole in which it resides can create a disturbance in the normal heat flow distribution, so the measurement may not be a good indication of the temperature that would exist at the same location in the absence of the thermocouple. Although these problems are not as severe with fast-response microthermocouples, the best use for embedded thermocouples is to measure bulk temperatures within the sliding bodies, and not flash temperatures. The bulk temperatures can be used effectively in determining boundary conditions for an analytical study or for calculating the distribution of frictional heat between the two contacting bodies (Berry and Barber, 1984). This is most easily done if temperatures are measured at several depths beneath the sliding surface, enabling the determination of heat flux values.

6.3.1.2 Dynamic Thermocouples

In the dynamic thermocouple technique, sometimes called the Herbert–Gottwein technique, a thermocouple junction is formed at the sliding interface by the contacting bodies themselves. It was originally developed to study contact temperatures at the interface between a cutting tool and workpiece during metal cutting (Shore, 1925). Later it was used to make measurements of surface temperature in a variety of sliding contacts (e.g., Cook and Bhushan, 1973). As long as the two contacting materials are dissimilar and produce a well-characterized thermal emf as a function of temperature, the two rubbing materials can be used as part of a thermocouple circuit. An example of the use of this technique is shown in Figure 6.21. In one typical application, a constantan ball formed one element of the thermocouple and the steel (iron) cylinder was the other (Furey, 1964). A measuring junction was formed wherever there was intimate contact between the ball and cylinder, and the measured temperature was the average temperature at the contact interface.

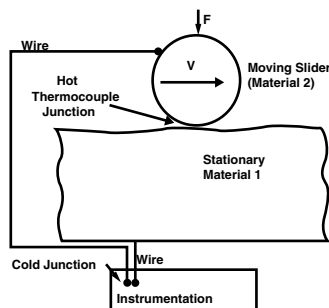


FIGURE 6.21 Schematic diagram of a dynamic thermocouple to measure temperature at contact between moving and stationary bodies made of dissimilar metals.

In a modification of the technique, a single thin wire of constantan (or similar material) could be embedded within the stationary sliding element, emerging at the contact surface. After insulating the wire from the surrounding material, using ceramic cement or epoxy, its end can be ground smooth with the remainder of the sliding surface. In that way, a dynamic thermocouple junction will be formed whenever the end of the wire is in contact with the other surface, and the contact location can be known with certainty (Kennedy, 1984).

Dynamic thermocouples have a very thin junction which consists of only the actual contact zone. As a result, they can respond very rapidly to changes in surface temperature. Because they can respond to flash temperatures as well as nominal contact temperatures, dynamic thermocouples have been found to give higher measurements than embedded thermocouples and faster transient response (Kennedy, 1984). The measurements are often lower than theoretical predictions, however (Furey, 1964), and questions persist about the accuracy and meaning of the thermal emf produced by the thermocouple. The emf is produced by a weighted average of all temperatures across the sliding thermocouple junction, and that average may differ from the peak contact temperature (T_c), especially when the thermocouple junction contains more than one contact spot and the junction size is changing with time (Shu et al., 1964). In addition, there is frequently some electrical noise generated at the contact interface, and the thermocouple output must be discriminated from this noise.

6.3.1.3 Thin Film Temperature Sensors

In recent years, microelectronic fabrication techniques, such as vapor deposition, have begun to be used to make surface temperature sensors. The advantage of using such techniques is that very small sensors, with rapid response time, can be formed on the surfaces.

The earliest vapor-deposited surface temperature sensors were thermistors used to measure surface temperatures on elastohydrodynamically lubricated components such as gear teeth. One such sensor consisted of a thin strip of titanium coated onto an alumina insulator on the surface of one of a pair of meshing teeth (Kannel and Bell, 1972). The resistance of the titanium strip is sensitive to temperature, so by monitoring the change in resistance the transient surface temperature changes could be measured. By nature the strip has a finite length and responds to all temperature changes along its length. Thus, it gives an integrated average measure of temperature and not a pointwise temperature measurement.

A recent embodiment of thin film fabrication techniques in surface temperature measurement has been the development of thin film thermocouples. Some of the earliest work on thin film thermocouples described the use of vapor deposition to produce thermocouple pairs from thin films of nickel and iron, copper and iron, copper and nickel, copper and constantan, and chromel and alumel (Marshall et al., 1966). More recently, similar techniques have been used successfully for measuring sliding surface temperatures (Tong et al., 1987; Schreck et al., 1992; Tian et al., 1992). The production of a thin film thermocouple (TFTC) involves the deposition of thin films (typically $<1\text{ }\mu\text{m}$ thick) of two different metals, such as nickel and copper, sandwiched between thin layers (also $<1\text{ }\mu\text{m}$ thick) of a hard, dielectric material such as Al_2O_3 . The measuring junction of the TFTC is deposited on the surface where frictional heat is generated. The dielectric layer beneath the thermocouple junction is necessary to electrically insulate the TFTC device from the underlying metallic surface. Owing to the softness of the thin metal films making up the thermocouple and connecting leads, it is also necessary to deposit a hard protective layer above the junction to limit damage to the TFTC device. The metal and dielectric films can be grown by physical vapor deposition techniques, with junction sizes as small as $10\text{ }\mu\text{m}^2$ or smaller and thicknesses less than $1\text{ }\mu\text{m}$. TFTC devices have been found to have extremely rapid ($<1\text{ }\mu\text{s}$) response to a sudden temperature change, and there have been reports of response as fast as 60 ns for Pt–Ir TFTC (Tong et al., 1987). It has also been shown that TFTC do not significantly disturb the heat flow from sliding contacts (Tian et al., 1992). Such sensors can measure the actual temperature of the contact interface, especially when the protective layer is very thin, and the small size of the measuring junctions enables them to respond rapidly to temperature changes at a specific point on the surface. If that point happens to be a flash temperature location, they can give a measure of maximum contact temperature (T_c). As with all thermocouples or thermistors, however, thin film devices cannot give a complete picture of surface

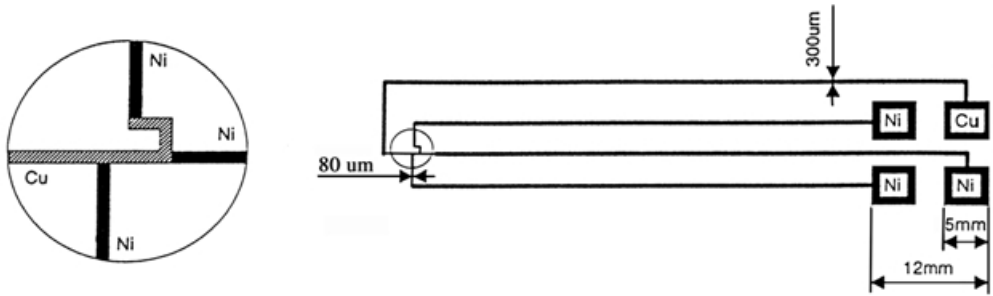


FIGURE 6.22 Thin film thermocouple with three measuring junctions. Line width = 80 μm at the thermocouple junctions.

TABLE 6.2 Representative Values of Total Emissivity of Solid Surfaces at 25°C

Material	Total Emissivity at 25°C
Copper, polished	0.03
Copper, oxidized	0.5
Iron, polished	0.08
Iron, oxidized	0.8
Carbon	0.8
Blackbody	1.0

Source: Adapted from Bedford, 1991.

temperature distribution, since that requires the measurement of temperature at a large number of points simultaneously. More recently, arrays of thin film thermocouples have been used to measure the temperature at multiple points on the surface simultaneously (Kennedy et al., 1997). This enables the determination of a portion of the surface temperature field in a sliding contact and can be useful in determining real contact area and pressure distribution. A typical thin film thermocouple array is shown in Figure 6.22. TFTC arrays have been developed with up to 64 thermocouple junctions in an area as small as 500 μm square.

6.3.2 Radiation Detection Techniques

Some of the most successful measurements of sliding contact temperatures have used techniques involving the detection of thermal radiation. It is well known that any surface with a temperature above absolute zero is a natural radiator of thermal energy. The radiation emitted by the object is a unique function of its temperature, with the Stefan–Boltzmann law (Equation 6.47) showing that much more power is emitted by a body at high temperatures than at low.

$$\Phi = \epsilon \sigma T^4 A \quad (6.47)$$

where Φ is the power (energy rate), T is the absolute temperature, σ is the Stefan–Boltzmann constant, A is the area of the heat source, and ϵ is the total emissivity of the surface emitting the radiation. ϵ is temperature-dependent and is also very dependent on the characteristics of the surface. Some typical emissivity values at room temperature are listed in Table 6.2.

The radiation is composed of photons of many wavelengths and, according to Planck's law, the monochromatic emissive power of a blackbody in a vacuum is:

$$w_{b,\lambda} = 2\pi Hc^2 \lambda^{-5} / (e^{hc/\lambda kT} - 1) \quad (6.48)$$

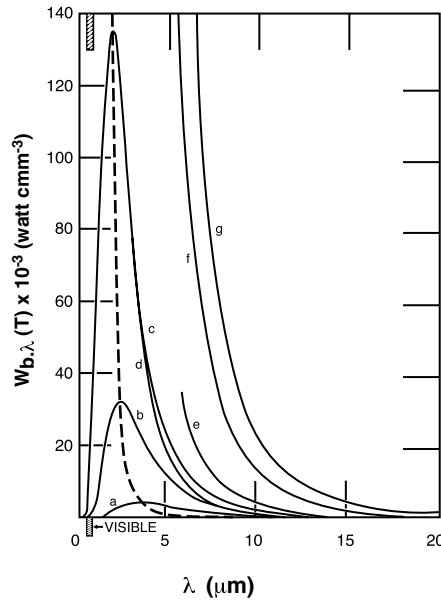


FIGURE 6.23 Spectral radiative emittance of a blackbody (Equation 6.48) at various temperatures: (a) 800 K, (b) 1200 K, (c) 1600 K, (d) 6000 K, (e) 10,000 K. (From Bedford, R.E. (1991), Blackbody radiation, in *Encyclopedia of Physics*, Lerner, R.G. and Trigg, G.L. (Eds.), VCH Publishers, Inc., New York, 104. Reprinted by permission of John Wiley & Sons, Inc.)

where λ is wavelength, c is the velocity of light in a vacuum, and H and K are Planck and Boltzmann constants, respectively. $w_{b,\lambda}$ is defined as the energy emitted per unit area at wavelength λ per unit wavelength in a small interval around λ . Integration of Equation 6.48 over all wavelengths leads to Equation 6.47 for the case of a blackbody ($\epsilon = 1$). Planck's law (Equation 6.48) is plotted in Figure 6.23 for some representative temperatures (Bedford, 1991). It can be noted that $w_{b,\lambda}$ is very low at small and long wavelengths, so most emissive power is found at wavelengths in the range $1 \mu\text{m} < \lambda < 10 \mu\text{m}$. For this reason, most successful attempts to measure temperature by detecting thermal radiation have concentrated on the infrared region of the spectrum (wavelengths of $0.75 \mu\text{m}$ to $500 \mu\text{m}$). If the surface temperature T is high enough, radiation in the visible part of the spectrum (400 nm to 750 nm) can also be detected. Several different radiation measurement techniques have been used with success in measuring surface temperatures, including photography, pyrometry, thermal imaging, and photon detection.

6.3.2.1 Optical and Infrared Photography

A photographic technique utilizing infrared-sensitive film was developed by Boothroyd (1961) for studying the temperature distribution in metal cutting. Similar methods have since been used in other studies of surface temperatures in machining, as well as for sliding components such as brakes (Santini and Kennedy, 1975) and in pin-on-disk sliding (Quinn and Winer, 1985). In most cases the camera is focused on the moving body as it emerges from a sliding contact, but successful photographs have also been made through a transparent window to a sapphire/metal or sapphire/ceramic contact (Quinn and Winer, 1985). Sapphire is a useful material for such studies because its mechanical and thermal properties are similar to those of steel and it is essentially transparent to radiation in the visible and near-infrared regions. A photograph showing hot spots on a tool steel pin sliding at high speed on a sapphire disk is shown in Figure 6.24 (Quinn and Winer, 1985). Temperatures of the spots were estimated to range from 950°C to 1200°C . The temperature distribution is best determined by measuring the optical density of the developed negative. The system must be calibrated to determine the density–temperature relationship of the film in the test configuration. This is usually accomplished by photographing specimens of the same material which had been heated to known temperatures and then comparing the optical density

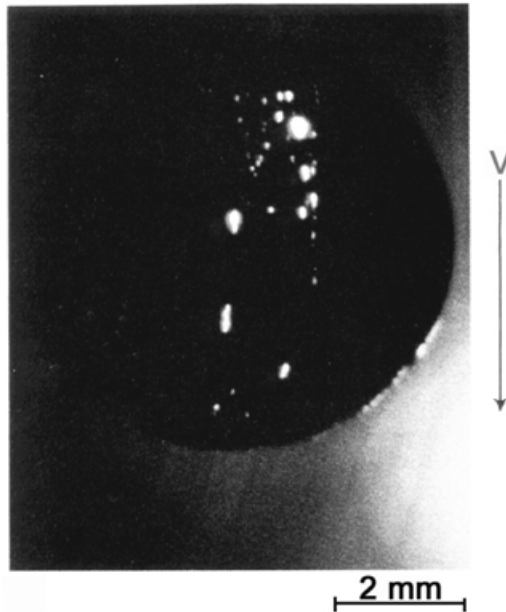


FIGURE 6.24 Photograph of hot spots on a steel pin (diameter = 6 mm) sliding at 2 m/s on a sapphire disk with a load of 26 N. Photo taken after 25 min of sliding. Exposure time = 1 s. (From Quinn, T.F.J. and Winer, W.O. (1985), *The thermal aspects of oxidational wear*, *Wear*, 102, 67-80. With permission.)

of the test film to that of the calibrated film. The same magnification and exposure time must be used in both calibration and test. When careful procedures are used, the method can give good indications of the surface temperature distribution on a sliding contact. In general, however, methods involving IR-sensitive film require exposure times (5 s or more) which are longer than the duration of flash temperatures, so they probably do not give a true indication of the highest temperature in a sliding contact. Shorter exposure times (<1 s) can be achieved with normal high-speed color film, as in [Figure 6.24](#), but such film operates with visible light, so is restricted to detection of high temperatures, perhaps 800°C or higher. IR film can respond to mean contact temperatures as low as 300°C, but even those temperatures occur only in rather severe sliding situations. An alternative is to use a modern infrared camera, which is essentially a scanning infrared detector. This will be described below.

6.3.2.2 Infrared Detectors

Infrared (IR) detection techniques have been widely used and improved since Parker and Marshall (1948) used an optical pyrometer to measure the surface temperature of a railroad wheel as it emerged from a brake shoe. Early pyrometers used the eye as a detector to match the brightness of the subject body with that of a standard lamp incorporated in the instrument. Improved models were later developed which employed a photoelectric detector in place of the eye. The detector essentially integrates Equation 6.48 over all wavelengths within its spectral range and over the surface area viewed by the detector. Since the temperature is generally not constant over the field of view, the detector output is a function of the average temperature over the area. In order to improve the accuracy of the temperature measurement and to approach a point measurement, most modern detectors are equipped with optics which limit the field of view to a small spot size, perhaps on the order of 100 to 500 μm diameter. The result is an infrared radiometric (IR) microscope, an example of which is shown schematically in [Figure 6.25](#). The IR microscope shown in [Figure 6.25](#) uses a liquid nitrogen-cooled detector made of indium antimonide (InSb), which has a spectral band of 2 to 5.6 μm . IR microscopes can measure transient temperature changes at a rate of up to 20 kHz or greater. They have been used effectively both with metallic components, where the detector can be focused on a spot just emerging from the contact zone (Griffioen et al., 1986), or

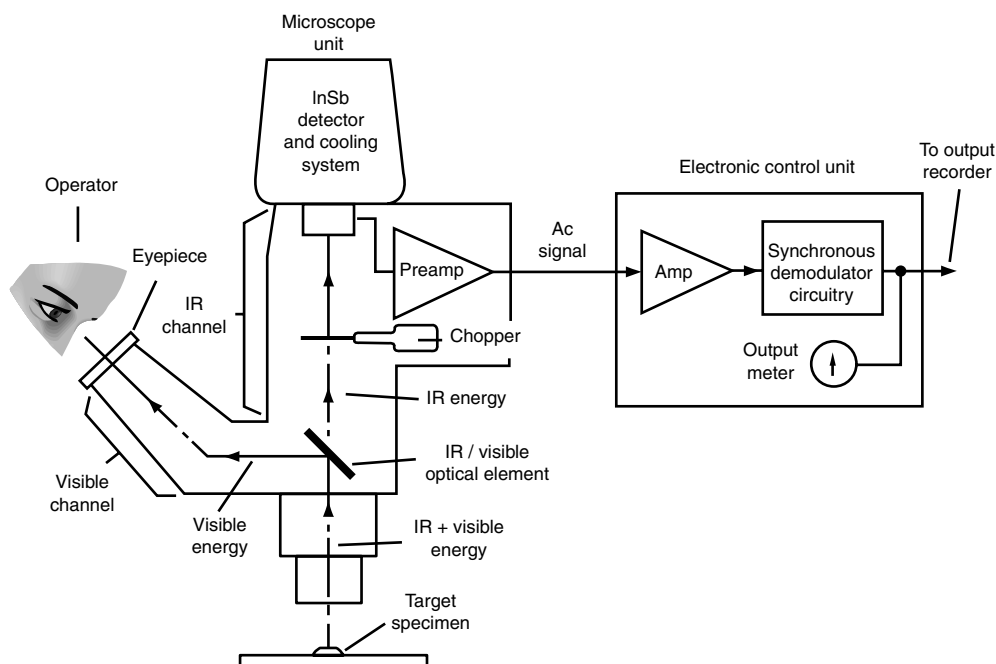


FIGURE 6.25 Schematic diagram of infrared radiometric microscope.

with a transparent sapphire component, in which case the detector would be focused through the sapphire onto the contact zone between sapphire and metal (Floquet and Play, 1981).

By limiting the field of view to a single small spot, the IR microscope can miss many contact events which occur at other spots within the area of contact. To overcome this limitation, investigators in recent years have been using a scanning infrared camera to study sliding surface temperatures (Griffioen et al., 1986; Gulino et al., 1986). A scanning IR camera, or infrared micro-imager, has a detector similar to that shown in Figure 6.25, but the detector is optically scanned over the contact surface in either of two modes, line scan or area scan. In the line scan mode, a fixed line, perhaps several mm in length, is scanned continuously at a rate of up to 2500 line scans per second. In the area scan mode, the rotation of a prism advances the line for each scan to produce a field consisting of typically 100 scan lines. The output is a video voltage map which is a function of the infrared radiation detected at that instant. The output from a complete area scan for the case of a silicon nitride pin sliding against a sapphire disk is shown in Figure 6.26.

It should be noted that, even at a scan rate of 2500 lines per second, it takes 40 ms to complete an area scan composed of 100 lines. This is considerably longer than the duration of flash temperatures, so it is unlikely that a field plot similar to Figure 6.26 is fully representative of the surface temperature distribution at any instant. It does, however, give a good indication of the contact conditions within the target area and the approximate temperatures reached at the hot spots. A better indication of transient temperatures at a given point can be achieved in the line scan mode, by continuously sweeping over the same line. Even in that mode, however, the transient times of the temperature fluctuations have been found to be less than the time required to complete a single line scan, and the flash temperature intervals may, in fact, be less than the 5 μ s or so between consecutive temperature measurements on the same scan line. Thus, measured contact temperatures may be less than actual flash temperatures, particularly if the hot spot is smaller than the detector's spot size and is very short-lived.

Methods have been devised to correct for the instantaneous temperature averaging that occurs within an IR detector (Griffioen et al., 1986), but those techniques are only approximate. As was stated earlier, an IR detector essentially integrates Equation 6.48, multiplied by the emissivity, over the area of the

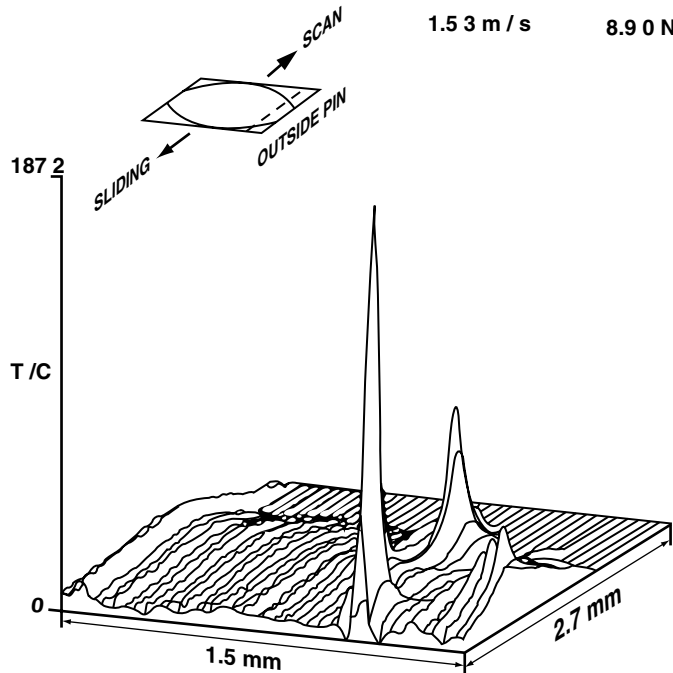


FIGURE 6.26 Temperature distribution on surface of silicon nitride pin in sliding contact with sapphire disk at velocity of 1.53 m/s and normal load of 8.9 N. Pin bulk temperature = 96°C. Area scan mode with infrared scanning camera. (From Griffioen, J.A., Bair, S., and Winer, W.O. (1986), Infrared surface temperature measurements in a sliding ceramic-ceramic contact, *Mechanisms and Surface Distress*, Dowson, D. et al. (Eds.), Butterworths, London, 238. With permission.)

detector's target spot and within its spectral range to get the equivalent of Equation 6.47. If a small hot spot whose temperature is desired is contained within a larger target spot, it is necessary to know the area of the spot so that its contribution to the summed detector output can be determined. Since hot spot areas are usually not known with certainty, the hot spot temperature may be inaccurately determined. A better technique, utilizing two separate detectors, was devised by Bair et al. (1991). If the emitted radiation is split between the two detectors and a different bandpass filter is placed in front of each detector, different values of radiated power will be measured at each of the two wavelengths, and each will be a function of two variables, hot spot area and temperature. The ratio of detected power at the two wavelengths can be used to determine the maximum temperature within the field of view (Bair et al., 1991). The hot spot area can also be determined, once its temperature has been calculated. The optical setup for this method is shown in Figure 6.27.

One factor that can lead to inaccuracies in temperature determination using any of the IR techniques is uncertainty about the emissivity of contacting surfaces during the sliding process. It is apparent from Table 6.2 that the total emissivity of a metallic (or nonmetallic) surface can vary considerably owing to oxidation, wear, or other changes in the surface characteristics. In order to get an accurate temperature reading from a radiating surface, an accurate value of emissivity must be known at that temperature. This can be accomplished by carefully determining the emissivity of reference surfaces similar to the contacting surfaces at temperatures throughout the range of interest. Methods can also be developed to handle the emissivity and transmissivity of any lubricant between the surfaces. Despite these procedures, emissivity remains an accuracy-limiting variable in many IR measurements of sliding surface temperatures, especially when the emissivity changes during the sliding process. These difficulties have been partially removed by the technique of Bair et al. (1991), for which the calculated temperature is independent of the emissivity as long as the spectral distribution of emissivity remains unchanged.

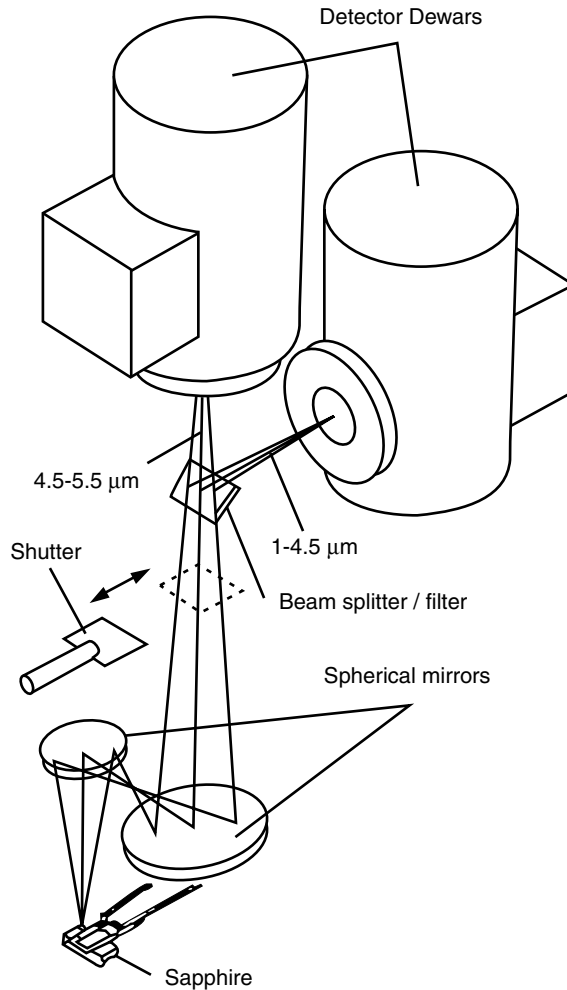


FIGURE 6.27 Optical arrangement for infrared temperature measurement system employing two detectors. (From Bair, S., Green, I., and Bhushan, B. (1991), Measurements of asperity temperatures of a read/write head slider bearing in hard magnetic recording disks, *ASME J. Tribol.*, 113, 547-554. With permission.)

There are several other limitations of infrared detectors when used to measure flash temperature rises. One is that if the size of the hot spot is smaller than the field of view of the detector, there will be a significant loss of accuracy in the temperature measurement. For current infrared detectors, the lower limit of hot spot size for which accurate measurements can be made is 1 to 2 μm (Chung and Wahl, 1992). Another potential limitation of infrared temperature measurement is that the time response (integration time) of the detector may be longer than the duration of the hot spot being measured. This can be a problem for small, rapidly moving hot spots.

As an alternative to infrared detectors, a surface temperature measurement method has been developed which uses a photomultiplier to collect photons emitted by a hot contact spot (Suzuki and Kennedy, 1991). Contact temperatures generally need to be a minimum of 400 to 500°C in order to get photons with enough energy to be detected by the photomultiplier, but the response time of the photomultiplier is very rapid (<30 ns). Therefore, the technique can be used for detecting flash temperatures of very short duration (2 μs or less), but it is not too useful for measuring mean contact temperatures, which are generally lower than 500°C. A further restriction is that the sliding test must be run in complete

darkness to eliminate noise which can dominate the output signal. This method is subject to a limitation similar to that of most infrared detectors, i.e., the area of the spot emitting the photons must be known in order to accurately determine the spot's temperature.

6.3.3 Ex Post Facto Methods

6.3.3.1 Metallographic Techniques

By examining the microstructure of sections of bodies which have undergone frictional heating, information can be gained about the temperatures the bodies witnessed in service. These techniques are generally dependent on the microstructural changes that occur as a result of the surface and near-surface temperatures. This change in microstructure can be detected after metallurgically sectioning the sliding body in a plane normal to the sliding direction. For some materials, etching of the near surface region of the cross-sectioned body can reveal a visible change in microstructure (Wright, 1978). For other materials, microhardness surveys have been found to be effective in determining the near-surface temperature distribution that the material had witnessed in service (Wright and Trent, 1973). In either case, the temperature contours are constructed by comparing the hardness or structural appearance variations with those of standard reference specimens that had been heat treated to known temperatures for known lengths of time (Wright, 1978).

Metallographic techniques generally require destruction of the sliding body for sectioning. Such post mortem investigations can give substantial information about what bulk surface and volumetric temperatures had occurred during earlier sliding, but they cannot provide instantaneous temperature measurements. They can only be used successfully with those materials which undergo a known change in microstructure or microhardness at the temperatures encountered in sliding. Another shortcoming is that microhardness and some microstructural changes can also be affected by plastic deformation in the contact region.

References

- Akkok, M., Ettles, C.M.M., and Calabrese, S.J. (1987), Parameters affecting the kinetic friction of ice, *ASME J. Tribol.*, 109, 552-561.
- Archard, J.F. (1958/59), The temperature of rubbing surfaces, *Wear*, 2, 438-455.
- Ashby, M.F., Abulawi, J., and Kong, H.S. (1991), Temperature maps for frictional heating in dry sliding, *Tribology Trans.*, 34, 577-587.
- Bair, S., Green, I., and Bhushan, B. (1991), Measurements of asperity temperatures of a read/write head slider bearing in hard magnetic recording disks, *ASME J. Tribol.*, 113, 547-554.
- Banerjee, B.N. and Burton, R.A. (1979), Experimental studies of thermoelastic effects in hydrodynamically lubricated face seals, *ASME J. Lubrication Technology*, 101, 275-282.
- Barber, J.R., Beamond, T.W., Waring, J.R., and Pritchard, C. (1986), Implications of thermoelastic instability for the design of brakes, *ASME J. Tribol.*, 107, 206-210.
- Bedford, R.E. (1991), Blackbody radiation, in *Encyclopedia of Physics*, Lerner, R.G. and Trigg, G.L. (Eds.), VCH Publishers, Inc., New York, 104.
- Berry, G.A. and Barber, J.R. (1984), The division of frictional heat — a guide to the nature of sliding contact, *ASME J. Tribol.*, 106, 405-415.
- Bhushan, B. (1987a), Magnetic head-media interface temperatures — part 1: analysis, *ASME J. Tribol.*, 109, 243-251.
- Bhushan, B. (1987b), Magnetic head-media interface temperatures — part 2: application to magnetic tapes, *ASME J. Tribol.*, 109, 252-256.
- Bhushan, B. (1992), Magnetic head-media interface temperatures — part 3: application to rigid disks, *ASME J. Tribol.*, 114, 420-430.
- Bhushan, B. (1999), *Principles and Applications of Tribology*, John Wiley & Sons, New York.

- Blok, H.A. (1937), Theoretical study of temperature rise at surfaces of actual contact under oiliness lubricating conditions, *Proc. General Discussion on Lubrication and Lubricants*, I.Mech.E., London, 222-235.
- Boothroyd, G. (1961), Photographic technique for the determination of metal cutting temperatures, *Br. J. Appl. Physics*, 12, 238.
- Bos, J. and Moes, H. (1994), Frictional heating of elliptic contacts, in *Dissipative Processes in Tribology*, Dowson, D. et al. (Eds.), Elsevier Science, Amsterdam, 491.
- Bos, J. and Moes, H. (1995), Frictional heating of tribological contacts, *ASME J. Tribol.*, 117, 171-177.
- Bowden, F.P. and Hughes, T.P. (1939), The mechanism of sliding on ice and snow, *Proc. R. Soc.*, A172, 280-298.
- Carignan, F.J. and Rabinowicz, E. (1980), Friction and wear at high sliding speeds, *ASLE Trans.*, 24, 451-459.
- Carslaw, H.S. and Jaeger, J.C. (1959), *Conduction of Heat in Solids*, 2nd ed., Clarendon Press, Oxford.
- Chung, Y-W. and Wahl, K.J. (1992), Fundamental limits of flash temperature measurements of asperities using infrared detectors, *Tribology Trans.*, 35, 447-450.
- Colin, F. and Floquet, A. (1986), Combination of finite element and integral transform techniques in a heat conduction quasi-static problem, *Int. J. Numer. Methods in Engg.*, 23, 13-26.
- Cook, N.H. and Bhushan, B. (1973), Sliding surface interface temperatures, *ASME J. Lubrication Technology*, 96, 59-64.
- Dow, T.A. (1980), Thermoelastic effects in brakes, *Wear*, 59, 213-221.
- Dow, T.A. and Kannel, J.W. (1982), Thermomechanical effects in high current density slip rings, *Wear*, 79, 93-105.
- Dyson, A. (1975), Scuffing — a review, *Tribology Int'l.*, 8, 77-87.
- Enthoven, J.C., Cann, P.M., and Spikes, H.A. (1993), Temperature and scuffing, *Tribology Trans.*, 36, 258-266.
- Ettles, C.M.M. (1986), The thermal control of friction at high sliding speeds, *ASME J. Tribol.*, 108, 98-104.
- Ettles, C.M.M. and Shen, J.H. (1988), The influence of frictional heating on the sliding friction of elastomers and polymers, *Rubber Chemistry and Technology*, 61, 119-136.
- Ettles, C.M.M., Dinc, O.S., and Calabrese, S.J. (1994), The effect of frictionally generated heat on lubricant transition, *Tribology Trans.*, 37, 420-424.
- Fein, R.S., Rowe, C.N., and Kreuze, K.L. (1959), Transition temperatures in sliding systems, *ASLE Trans.*, 2, 50-57.
- Floquet, A., Play, D., and Godet, M. (1977), Surface temperatures in distributed contacts: application to bearing design, *ASME J. Lubrication Technology*, 99, 277-283.
- Floquet, A. and Play, D. (1981), Contact temperatures in dry bearings. Three-dimensional theory and verification, *ASME J. Tribol.*, 103, 243-252.
- Floquet, A. (1983), Thermal considerations in the design of tribometers, *Wear*, 88, 45-56.
- Furey, M.J. (1964), Surface temperatures in sliding contact, *ASLE Trans.*, 7, 133-146.
- Gecim, B. and Winer, W.O. (1985), Transient temperatures in the vicinity of an asperity contact, *ASME J. Tribol.*, 107, 333-342.
- Greenwood, J.A. (1991), An interpolation formula for flash temperatures, *Wear*, 150, 153-158.
- Griffioen, J.A., Bair, S., and Winer, W.O. (1986), Infrared surface temperature measurements in a sliding ceramic-ceramic contact, *Mechanisms and Surface Distress*, Dowson, D. et al. (Eds.), Butterworths, London, 238.
- Gulino, R., Bair, S., Winer, W.O., and Bhushan, B. (1986), Temperature measurements of microscopic areas within a simulated head/tape interface using infrared radiometric technique, *ASME J. Tribol.*, 108, 29-34.
- Hou, Z-B. and Komanduri, R. (1998), Magnetic field assisted finishing of ceramics — part 1: thermal model, *ASME J. Tribol.*, 120, 645-651.
- Hsu, S.M. and Shen, M.C. (1996), Ceramic wear maps, *Wear*, 200, 154-175.
- Jacobsen, B. (1990), Mixed lubrication, *Wear*, 136, 99-116.

- Jaeger, J.C. (1942), Moving sources of heat and the temperature of sliding contacts, *Proc. R. Soc. NSW*, 76, 203-224.
- Ju, F.D. and Huang, J.H. (1982), Heat checking in the contact zone of a bearing seal, *Wear*, 79, 107-118.
- Kannel, J.W. and Bell, J.C. (1972), A method for estimation of temperatures in lubricated rolling-sliding gear or bearing elastohydrodynamic contacts, *Proc. I.Mech.E. EHD Symposium*, London, 118-130.
- Kennedy, F.E. (1981), Surface temperatures in sliding systems — a finite element analysis, *ASME J. Lubrication Technology*, 103, 90-96.
- Kennedy, F.E. (1982), Single-pass rub phenomena — analysis and experiment, *ASME J. Lubrication Technology*, 104, 582-588.
- Kennedy, F.E. (1984), Thermal and thermomechanical effects in dry sliding, *Wear*, 100, 453-476.
- Kennedy, F.E. (1992), Surface temperature measurement, *Friction, Lubrication and Wear Technology, Metals Handbook, 4, 10th ed.*, Blau, P.J. (Ed.), ASM International, Metals Park, OH, 438.
- Kennedy, F.E. and Hussaini, S.Z. (1987), Thermo-mechanical analysis of dry sliding systems, *Computers and Structures*, 26, 345-355.
- Kennedy, F.E. and Karpe, S.A. (1982), Thermocracking of a mechanical face seal, *Wear*, 79, 21-36.
- Kennedy, F.E. and Tian, X. (1994), The effect of interfacial temperature on friction and wear of thermoplastics in the thermal control regime, in *Dissipative Processes in Tribology* Dowson, D. et al. (Eds.), Elsevier Science, Amsterdam, 235.
- Kennedy, F.E., Colin, F., Floquet, A., and Glovsky, R. (1984), Improved techniques for finite element analysis of sliding surface temperatures, *Developments in Numerical and Experimental Methods Applied to Tribology*, Dowson, D. et al. (Eds.), Butterworths, London, 138.
- Kennedy, F.E., Frusescu, D., and Li, J. (1997), Thin film thermocouple arrays for sliding surface temperature measurement, *Wear*, 207, 46-54.
- Kennedy, F.E., Schulson, E.M., and Jones, D.E. (2000), The friction of ice on ice at low sliding velocities, *Phil. Mag. A*, 80, 1093-1110.
- Kuhlmann-Wilsdorf, D. (1987), Temperatures at interfacial contact spots: dependence on velocity and on role reversal of two materials in sliding contact, *ASME J. Tribol.*, 109, 321-329.
- Lancaster, J.K. (1971), Estimation of the limiting PV relationships for thermoplastic bearing materials, *Tribology*, 4, 82-86.
- Landman, U., Luedtke, W.D., and Ringer, E.M. (1993), Molecular dynamics simulations of adhesive contact formation and friction, in *Fundamentals of Friction: Macroscopic and Microscopic Processes*, Singer, I.L. and Pollock, H.M. (Eds.), Kluwer Academic Publishers, Dordrecht, 463.
- Lee, S.W., Hsu, S.M., and Shen, M.C. (1993), Ceramic wear maps: zirconia, *J. Am. Ceram. Soc.*, 76, 1937.
- Lim, S.C. and Ashby, M.F. (1987), Wear-mechanism maps, *Acta Metall.*, 35, 1-24.
- Ling, F.F. and Saibel, E. (1957/58), Thermal aspects of galling of dry metallic surfaces in sliding contact, *Wear*, 1, 80-91.
- Ling, F.F. (1959), A quasi-iterative method for computing interface temperature distributions, *Z Agnew. Math. Physik.*, X, 461-474.
- Ling, F.F. and Simkins, T.E. (1963), Measurement of pointwise junction condition of temperature at the interface of two bodies in sliding contact, *ASME J. Basic Eng.*, 85, 481-487.
- Ling, F.F. (1973), *Surface Mechanics*, John Wiley & Sons, New York.
- Marshall, R., Atlas, L., and Putner, T. (1966), The preparation and performance of thin thermocouples, *J. Sci. Instrum.*, 43, p. 144.
- Montgomery, R.S. (1976), Friction and wear at high sliding speeds, *ASLE Transactions*, 36, 275-298.
- Oksanen, P. and Keinonen, J. (1982), The mechanism of friction of ice, *Wear*, 78, 315-324.
- Parker, R.C. and Marshall, P.R. (1948), The measurement of the temperature of sliding surfaces with particular reference to railway blocks, *Proc., Inst. Mech. Eng. London*, 158, 209-229.
- Quinn, T.F.J. (1983), Review of oxidational wear, *Tribology Intl.*, 16, 257-271.
- Quinn, T.F.J. and Winer, W.O. (1985), The thermal aspects of oxidational wear, *Wear*, 102, 67-80.
- Reed, R.P. (1982), Thermoelectric thermometry: a functional model, in *Temperature — Its Measurement and Control in Science and Industry*, Schooley, J.F. (Ed.), American Inst. of Physics, New York, 5, 915.

- Rigney, D.A. and Hirth, J.P. (1979), Plastic deformation and sliding friction of metals, *Wear*, 53, 345-370.
- Rozeanu, L. and Pnueli, D. (1978), Two temperature gradients models for friction failure, *ASME J. Lubrication Technology* 100, 479-484.
- Santini, J.J. and Kennedy, F. E. (1975), An experimental investigation of surface temperatures and wear in disk brakes, *Lubr. Eng.*, 31, 402-417.
- Schreck, E., Fontana, R.E., and Singh, G.P. (1992), Thin film thermocouple sensors for measurement of contact temperatures during slider asperity interactions on magnetic recording disks, *IEEE Trans. on Magnetics*, 28, 2548-2550.
- Shore, H. (1925), Thermoelectric measurement of cutting tool temperature, *J. Wash. Acad. Sci.*, 15, 85-88.
- Shu, H.H., Gaylord, E.W., and Hughes, W.F. (1964), The relation between the rubbing interface temperature distribution and dynamic thermocouple temperature, *ASME J. Basic Engg.*, 86, 417-422.
- Spikes, H.A. and Cameron, A. (1974), Additive interference in dibenzyl disulphide extreme lubrication, *ASLE Trans.*, 16, 283-289.
- Stachowiak, G.W. and Batchelor, A.W. (1993), *Engineering Tribology*, Elsevier, Amsterdam.
- Suzuki, S. and Kennedy, F.E. (1988), Friction and temperature at head-disk interface in contact start/stop tests, *Tribology and Mechanics of Magnetic Storage Systems*, 5, 30-36.
- Suzuki, S. and Kennedy, F.E. (1991), The detection of flash temperatures in a sliding contact by the method of tribo-induced thermoluminescence, *ASME J. Tribol.*, 113, 120-127.
- Tian, X., Kennedy, F.E., Deacutis, J.J., and Henning, A.K. (1992), The development and use of thin film thermocouples for contact temperature measurement, *Tribology Trans.*, 35, 491-499.
- Tian, X. and Kennedy, F.E. (1993), Contact surface temperature models for finite bodies in dry and boundary lubricated sliding, *ASME J. Tribol.*, 115, 411-418.
- Tian, X. and Kennedy, F.E. (1994), Maximum and average flash temperatures in sliding contacts, *ASME J. Tribol.*, 116, 167-174.
- Ting, B.Y. (1988), Thermomechanical wear theory, Ph.D. dissertation, Georgia Institute of Technology, Atlanta, GA.
- Tong, H.M., Arjavalasingam, G., Haynes, R.D., Hyer, G.N., and Ritsko, J.J. (1987), High temperature thin-film Pt-Ir thermocouple with fast time response, *Re Sci. Instrum.*, 58, 875.
- Uetz, H. and Föhl, J. (1978), Wear as an energy transformation process, *Wear*, 49, 253-264.
- Wright, P.K. and Trent, E.M. (1973), Metallographic methods of determining temperature gradients in cutting tools, *J. Iron Steel Inst.*, 211, 364-388.
- Wright, P.K. (1978), Correlation of tempering effects with temperature distribution in steel cutting tools, *ASME J. Engineering for Industry*, 100, 131-136.
- Yevtushenko, A.A., Ivanyk, E.G., and Ukhanska, O.M. (1997) Transient temperature of local moving areas of sliding contact, *Tribology Intl.*, 30, 209-214.